

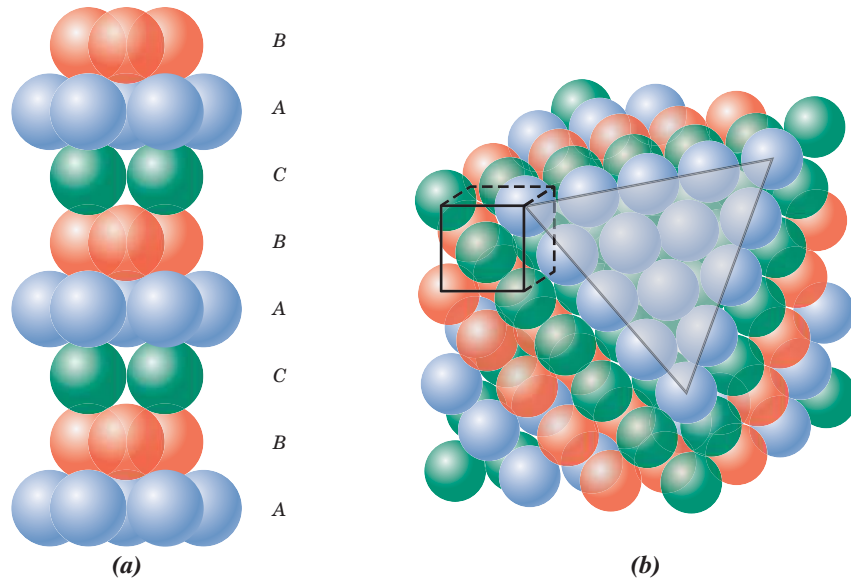


Figure 3.17 (a) Close-packed stacking sequence for the face-centered cubic structure. (b) A corner has been removed to show the relation between the stacking of close-packed planes of atoms and the FCC crystal structure; the heavy triangle outlines a (111) plane.

[Figure (b) adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.]

crystallographic planes and directions in terms of directional and Miller indices; furthermore, on occasion it is important to ascertain the atomic and ionic arrangements of particular crystallographic planes. Also, the crystal structures of a number of ceramic materials may be generated by the stacking of close-packed planes of ions (Section 12.2).

Crystalline and Noncrystalline Materials

3.13 SINGLE CRYSTALS

single crystal

For a crystalline solid, when the periodic and repeated arrangement of atoms is perfect or extends throughout the entirety of the specimen without interruption, the result is a **single crystal**. All unit cells interlock in the same way and have the same orientation. Single crystals exist in nature, but they can also be produced artificially. They are ordinarily difficult to grow because the environment must be carefully controlled.

If the extremities of a single crystal are permitted to grow without any external constraint, the crystal assumes a regular geometric shape having flat faces, as with some of the gemstones; the shape is indicative of the crystal structure. An iron pyrite single crystal is shown in Figure 3.18. Within the past few years, single crystals have become extremely important in many modern technologies, in particular electronic microcircuits, which employ single crystals of silicon and other semiconductors.

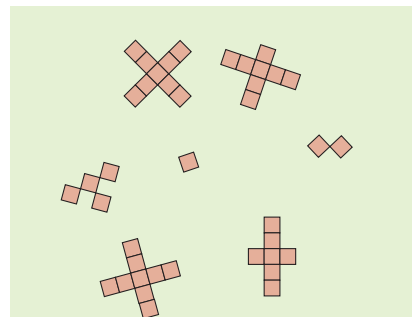
3.14 POLYCRYSTALLINE MATERIALS

grain polycrystalline

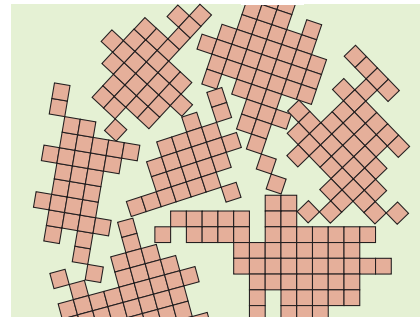
Most crystalline solids are composed of a collection of many small crystals or **grains**; such materials are termed **polycrystalline**. Various stages in the solidification of a polycrystalline specimen are represented schematically in Figure 3.19. Initially, small



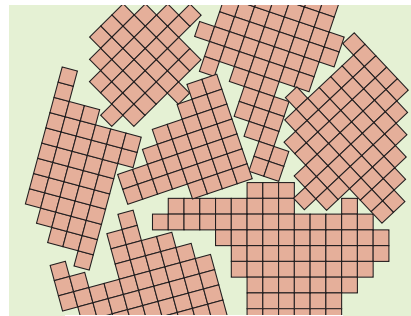
Figure 3.18 An iron pyrite single crystal that was found in Navajún, La Rioja, Spain.



(a)



(b)



(c)



(d)

Figure 3.19 Schematic diagrams of the various stages in the solidification of a polycrystalline material; the square grids depict unit cells. (a) Small crystallite nuclei. (b) Growth of the crystallites; the obstruction of some grains that are adjacent to one another is also shown. (c) Upon completion of solidification, grains having irregular shapes have formed. (d) The grain structure as it would appear under the microscope; dark lines are the grain boundaries. (Adapted from W. Rosenhain, *An Introduction to the Study of Physical Metallurgy*, 2nd edition, Constable & Company Ltd., London, 1915.)

crystals or nuclei form at various positions. These have random crystallographic orientations, as indicated by the square grids. The small grains grow by the successive addition from the surrounding liquid of atoms to the structure of each. The extremities of adjacent grains impinge on one another as the solidification process approaches completion. As indicated in Figure 3.19, the crystallographic orientation varies from grain to grain. Also, there exists some atomic mismatch within the region where two grains meet; this area, called a **grain boundary**, is discussed in more detail in Section 4.6.

grain boundary

3.15 ANISOTROPY

The physical properties of single crystals of some substances depend on the crystallographic direction in which measurements are taken. For example, the elastic modulus, the electrical conductivity, and the index of refraction may have different values in the [100] and [111] directions. This directionality of properties is termed **anisotropy**, and it is associated with the variance of atomic or ionic spacing with crystallographic direction. Substances in which measured properties are independent of the direction of measurement are **isotropic**. The extent and magnitude of anisotropic effects in crystalline materials are functions of the symmetry of the crystal structure; the degree of anisotropy increases with decreasing structural symmetry—triclinic structures normally are highly anisotropic. The modulus of elasticity values at [100], [110], and [111] orientations for several metals are presented in Table 3.4.

anisotropy

isotropic

For many polycrystalline materials, the crystallographic orientations of the individual grains are totally random. Under these circumstances, even though each grain may be anisotropic, a specimen composed of the grain aggregate behaves isotropically. Also, the magnitude of a measured property represents some average of the directional values. Sometimes the grains in polycrystalline materials have a preferential crystallographic orientation, in which case the material is said to have a “texture.”

The magnetic properties of some iron alloys used in transformer cores are anisotropic—that is, grains (or single crystals) magnetize in a $\langle 100 \rangle$ -type direction easier than any other crystallographic direction. Energy losses in transformer cores are minimized by utilizing polycrystalline sheets of these alloys into which have been introduced a *magnetic texture*: most of the grains in each sheet have a $\langle 100 \rangle$ -type crystallographic direction that is aligned (or almost aligned) in the same direction, which is oriented parallel to the direction of the applied magnetic field. Magnetic textures for iron alloys are discussed in detail in the Material of Importance box in Chapter 20 following Section 20.9.

Table 3.4

Modulus of Elasticity
Values for Several
Metals at Various
Crystallographic
Orientations

Metal	Modulus of Elasticity (GPa)		
	[100]	[110]	[111]
Aluminum	63.7	72.6	76.1
Copper	66.7	130.3	191.1
Iron	125.0	210.5	272.7
Tungsten	384.6	384.6	384.6

Source: R. W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 3rd edition. Copyright © 1989 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.

3.16 X-RAY DIFFRACTION: DETERMINATION OF CRYSTAL STRUCTURES

Historically, much of our understanding regarding the atomic and molecular arrangements in solids has resulted from x-ray diffraction investigations; furthermore, x-rays are still very important in developing new materials. We now give a brief overview of the diffraction phenomenon and how, using x-rays, atomic interplanar distances and crystal structures are deduced.

The Diffraction Phenomenon

Diffraction occurs when a wave encounters a series of regularly spaced obstacles that (1) are capable of scattering the wave, and (2) have spacings that are comparable in magnitude to the wavelength. Furthermore, diffraction is a consequence of specific phase relationships established between two or more waves that have been scattered by the obstacles.

Consider waves 1 and 2 in Figure 3.20a, which have the same wavelength (λ) and are in phase at point $O-O'$. Now let us suppose that both waves are scattered

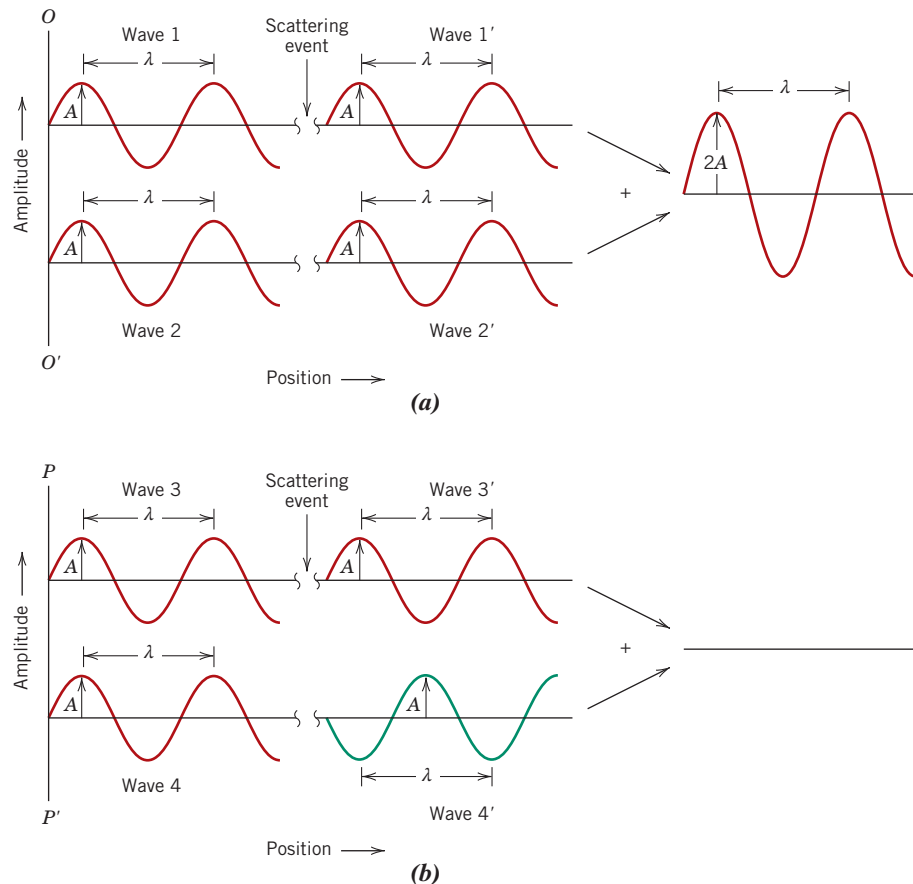


Figure 3.20 (a) Demonstration of how two waves (labeled 1 and 2) that have the same wavelength λ and remain in phase after a scattering event (waves 1' and 2') constructively interfere with one another. The amplitudes of the scattered waves add together in the resultant wave. (b) Demonstration of how two waves (labeled 3 and 4) that have the same wavelength and become out of phase after a scattering event (waves 3' and 4') destructively interfere with one another. The amplitudes of the two scattered waves cancel one another.

diffraction

in such a way that they traverse different paths. The phase relationship between the scattered waves, which depends upon the difference in path length, is important. One possibility results when this path length difference is an integral number of wavelengths. As noted in Figure 3.20a, these scattered waves (now labeled 1' and 2') are still in phase. They are said to mutually reinforce (or constructively interfere with) one another; when amplitudes are added, the wave shown on the right side of the figure results. This is a manifestation of **diffraction**, and we refer to a *diffracted beam* as one composed of a large number of scattered waves that mutually reinforce one another.

Other phase relationships are possible between scattered waves that will not lead to this mutual reinforcement. The other extreme is that demonstrated in Figure 3.20b, in which the path length difference after scattering is some integral number of *half*-wavelengths. The scattered waves are out of phase—that is, corresponding amplitudes cancel or annul one another, or destructively interfere (i.e., the resultant wave has zero amplitude), as indicated on the right side of the figure. Of course, phase relationships intermediate between these two extremes exist, resulting in only partial reinforcement.

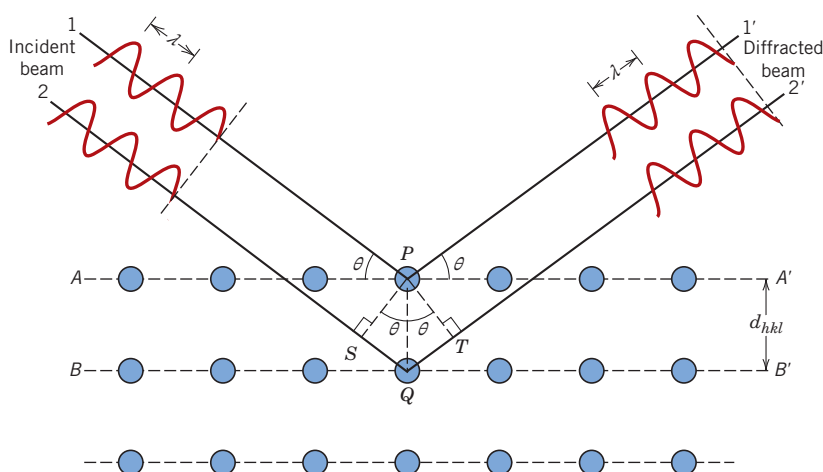
X-Ray Diffraction and Bragg's Law

X-rays are a form of electromagnetic radiation that have high energies and short wavelengths—wavelengths on the order of the atomic spacings for solids. When a beam of x-rays impinges on a solid material, a portion of this beam is scattered in all directions by the electrons associated with each atom or ion that lies within the beam's path. Let us now examine the necessary conditions for diffraction of x-rays by a periodic arrangement of atoms.

Consider the two parallel planes of atoms $A-A'$ and $B-B'$ in Figure 3.21, which have the same h , k , and l Miller indices and are separated by the interplanar spacing d_{hkl} . Now assume that a parallel, monochromatic, and coherent (in-phase) beam of x-rays of wavelength λ is incident on these two planes at an angle θ . Two rays in this beam, labeled 1 and 2, are scattered by atoms P and Q . Constructive interference of the scattered rays 1' and 2' occurs also at an angle θ to the planes if the path length difference between 1– P –1' and 2– Q –2' (i.e., $\overline{SQ} + \overline{QT}$) is equal to a whole number, n , of wavelengths—that is, the condition for diffraction is

$$n\lambda = \overline{SQ} + \overline{QT} \quad (3.19)$$

Figure 3.21 Diffraction of x-rays by planes of atoms ($A-A'$ and $B-B'$).



Bragg's law—
relationship among
x-ray wavelength,
interplanar spacing,
and angle of
diffraction for
constructive
interference

Bragg's law

or

$$\begin{aligned} n\lambda &= d_{hkl} \sin \theta + d_{hkl} \sin \theta \\ &= 2d_{hkl} \sin \theta \end{aligned} \quad (3.20)$$

Equation 3.20 is known as **Bragg's law**; n is the order of reflection, which may be any integer (1, 2, 3, . . .) consistent with $\sin \theta$ not exceeding unity. Thus, we have a simple expression relating the x-ray wavelength and interatomic spacing to the angle of the diffracted beam. If Bragg's law is not satisfied, then the interference will be nonconstructive so as to yield a very low-intensity diffracted beam.

The magnitude of the distance between two adjacent and parallel planes of atoms (i.e., the interplanar spacing d_{hkl}) is a function of the Miller indices (h , k , and l) as well as the lattice parameter(s). For example, for crystal structures that have cubic symmetry,

Interplanar spacing
for a plane having
indices h , k , and l

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad (3.21)$$

in which a is the lattice parameter (unit cell edge length). Relationships similar to Equation 3.21, but more complex, exist for the other six crystal systems noted in Table 3.2.

Bragg's law, Equation 3.20, is a necessary but not sufficient condition for diffraction by real crystals. It specifies when diffraction will occur for unit cells having atoms positioned only at cell corners. However, atoms situated at other sites (e.g., face and interior unit cell positions as with FCC and BCC) act as extra scattering centers, which can produce out-of-phase scattering at certain Bragg angles. The net result is the absence of some diffracted beams that, according to Equation 3.20, should be present. Specific sets of crystallographic planes that do not give rise to diffracted beams depend on crystal structure. For the BCC crystal structure, $h + k + l$ must be even if diffraction is to occur, whereas for FCC, h , k , and l must all be either odd or even; diffracted beams for all sets of crystallographic planes are present for the simple cubic crystal structure (Figure 3.3). These restrictions, called *reflection rules*, are summarized in Table 3.5.¹²



Concept Check 3.3 For cubic crystals, as values of the planar indices h , k , and l increase, does the distance between adjacent and parallel planes (i.e., the interplanar spacing) increase or decrease? Why?

[The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

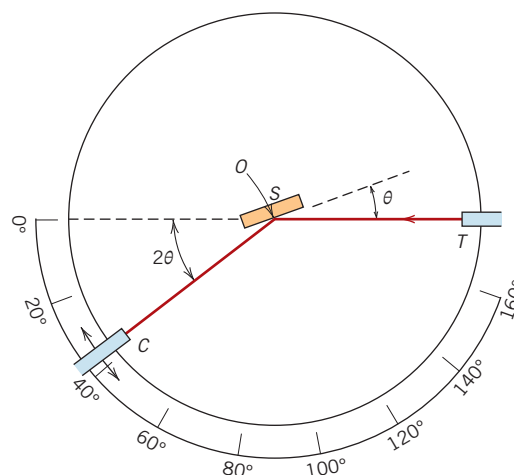
Table 3.5

X-Ray Diffraction
Reflection Rules and
Reflection Indices
for Body-Centered
Cubic, Face-Centered
Cubic, and Simple
Cubic Crystal
Structures

Crystal Structure	Reflections Present	Reflection Indices for First Six Planes
BCC	$(h + k + l)$ even	110, 200, 211, 220, 310, 222
FCC	h , k , and l either all odd or all even	111, 200, 220, 311, 222, 400
Simple cubic	All	100, 110, 111, 200, 210, 211

¹²Zero is considered to be an even integer.

Figure 3.22 Schematic diagram of an x-ray diffractometer; T = x-ray source, S = specimen, C = detector, and O = the axis around which the specimen and detector rotate.



Diffraction Techniques

One common diffraction technique employs a powdered or polycrystalline specimen consisting of many fine and randomly oriented particles that are exposed to monochromatic x-radiation. Each powder particle (or grain) is a crystal, and having a large number of them with random orientations ensures that some particles are properly oriented such that every possible set of crystallographic planes will be available for diffraction.

The *diffractometer* is an apparatus used to determine the angles at which diffraction occurs for powdered specimens; its features are represented schematically in Figure 3.22. A specimen S in the form of a flat plate is supported so that rotations about the axis labeled O are possible; this axis is perpendicular to the plane of the page. The monochromatic x-ray beam is generated at point T , and the intensities of diffracted beams are detected with a counter labeled C in the figure. The specimen, x-ray source, and counter are coplanar.

The counter is mounted on a movable carriage that may also be rotated about the O axis; its angular position in terms of 2θ is marked on a graduated scale.¹³ Carriage and specimen are mechanically coupled such that a rotation of the specimen through θ is accompanied by a 2θ rotation of the counter; this ensures that the incident and reflection angles are maintained equal to one another (Figure 3.22). Collimators are incorporated within the beam path to produce a well-defined and focused beam. Utilization of a filter provides a near-monochromatic beam.

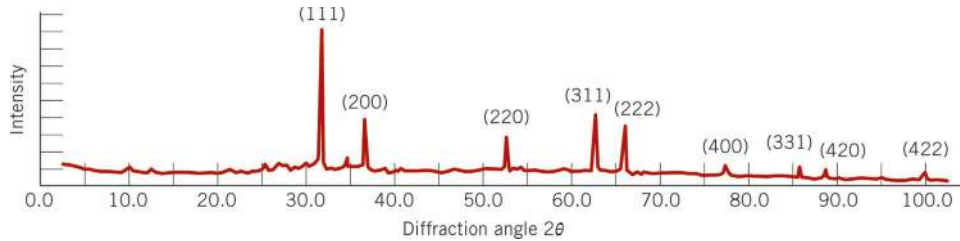
As the counter moves at constant angular velocity, a recorder automatically plots the diffracted beam intensity (monitored by the counter) as a function of 2θ ; 2θ is termed the *diffraction angle*, which is measured experimentally. Figure 3.23 shows a diffraction pattern for a powdered specimen of lead. The high-intensity peaks result when the Bragg diffraction condition is satisfied by some set of crystallographic planes. These peaks are plane-indexed in the figure.

Other powder techniques have been devised in which diffracted beam intensity and position are recorded on a photographic film instead of being measured by a counter.

¹³Note that the symbol θ has been used in two different contexts for this discussion. Here, θ represents the angular locations of both x-ray source and counter relative to the specimen surface. Previously (e.g., Equation 3.20), it denoted the angle at which the Bragg criterion for diffraction is satisfied.

Figure 3.23

Diffraction pattern for powdered lead.



One of the primary uses of x-ray diffractometry is for the determination of crystal structure. The unit cell size and geometry may be resolved from the angular positions of the diffraction peaks, whereas the arrangement of atoms within the unit cell is associated with the relative intensities of these peaks.

X-rays, as well as electron and neutron beams, are also used in other types of material investigations. For example, crystallographic orientations of single crystals are possible using x-ray diffraction (or Laue) photographs. The chapter-opening photograph (*a*) was generated using an incident x-ray beam that was directed on a magnesium crystal; each spot (with the exception of the darkest one near the center) resulted from an x-ray beam that was diffracted by a specific set of crystallographic planes. Other uses of x-rays include qualitative and quantitative chemical identifications and the determination of residual stresses and crystal size.

EXAMPLE PROBLEM 3.13

Interplanar Spacing and Diffraction Angle Computations

For BCC iron, compute (a) the interplanar spacing and (b) the diffraction angle for the (220) set of planes. The lattice parameter for Fe is 0.2866 nm. Assume that monochromatic radiation having a wavelength of 0.1790 nm is used, and the order of reflection is 1.

Solution

- (a) The value of the interplanar spacing d_{hkl} is determined using Equation 3.21, with $a = 0.2866$ nm, and $h = 2$, $k = 2$, and $l = 0$ because we are considering the (220) planes. Therefore,

$$\begin{aligned} d_{hkl} &= \frac{a}{\sqrt{h^2 + k^2 + l^2}} \\ &= \frac{0.2866 \text{ nm}}{\sqrt{(2)^2 + (2)^2 + (0)^2}} = 0.1013 \text{ nm} \end{aligned}$$

- (b) The value of θ may now be computed using Equation 3.20, with $n = 1$ because this is a first-order reflection:

$$\begin{aligned} \sin \theta &= \frac{n\lambda}{2d_{hkl}} = \frac{(1)(0.1790 \text{ nm})}{(2)(0.1013 \text{ nm})} = 0.884 \\ \theta &= \sin^{-1}(0.884) = 62.13^\circ \end{aligned}$$

The diffraction angle is 2θ , or

$$2\theta = (2)(62.13^\circ) = 124.26^\circ$$

EXAMPLE PROBLEM 3.14**Interplanar Spacing and Lattice Parameter Computations for Lead**

Figure 3.23 shows an x-ray diffraction pattern for lead taken using a diffractometer and monochromatic x-radiation having a wavelength of 0.1542 nm; each diffraction peak on the pattern has been indexed. Compute the interplanar spacing for each set of planes indexed; also, determine the lattice parameter of Pb for each of the peaks. For all peaks, assume the order of diffraction is 1.

Solution

For each peak, in order to compute the interplanar spacing and the lattice parameter we must employ Equations 3.20 and 3.21, respectively. The first peak of Figure 3.23, which results from diffraction by the (111) set of planes, occurs at $2\theta = 31.3^\circ$; the corresponding interplanar spacing for this set of planes, using Equation 3.20, is equal to

$$d_{111} = \frac{n\lambda}{2 \sin \theta} = \frac{(1)(0.1542 \text{ nm})}{(2) \left[\sin \left(\frac{31.3^\circ}{2} \right) \right]} = 0.2858 \text{ nm}$$

And, from Equation 3.21, the lattice parameter a is determined as

$$\begin{aligned} a &= d_{hkl} \sqrt{h^2 + k^2 + l^2} \\ &= d_{111} \sqrt{(1)^2 + (1)^2 + (1)^2} \\ &= (0.2858 \text{ nm}) \sqrt{3} = 0.4950 \text{ nm} \end{aligned}$$

Similar computations are made for the next four peaks; the results are tabulated below:

Peak Index	2θ	$d_{hkl} \text{ (nm)}$	$a \text{ (nm)}$
200	36.6	0.2455	0.4910
220	52.6	0.1740	0.4921
311	62.5	0.1486	0.4929
222	65.5	0.1425	0.4936

3.17 NONCRYSTALLINE SOLIDS**noncrystalline****amorphous**

It has been mentioned that **noncrystalline** solids lack a systematic and regular arrangement of atoms over relatively large atomic distances. Sometimes such materials are also called **amorphous** (meaning literally “without form”), or supercooled liquids, inasmuch as their atomic structure resembles that of a liquid.

An amorphous condition may be illustrated by comparison of the crystalline and noncrystalline structures of the ceramic compound silicon dioxide (SiO_2), which may exist in both states. Figures 3.24a and 3.24b present two-dimensional schematic diagrams for both structures of SiO_2 . Even though each silicon ion bonds to three oxygen ions (and a fourth oxygen ion above the plane) for both states, beyond this, the structure is much more disordered and irregular for the noncrystalline structure.

Whether a crystalline or an amorphous solid forms depends on the ease with which a random atomic structure in the liquid can transform to an ordered state during solidification. Amorphous materials, therefore, are characterized by atomic or molecular structures that are relatively complex and become ordered only with some

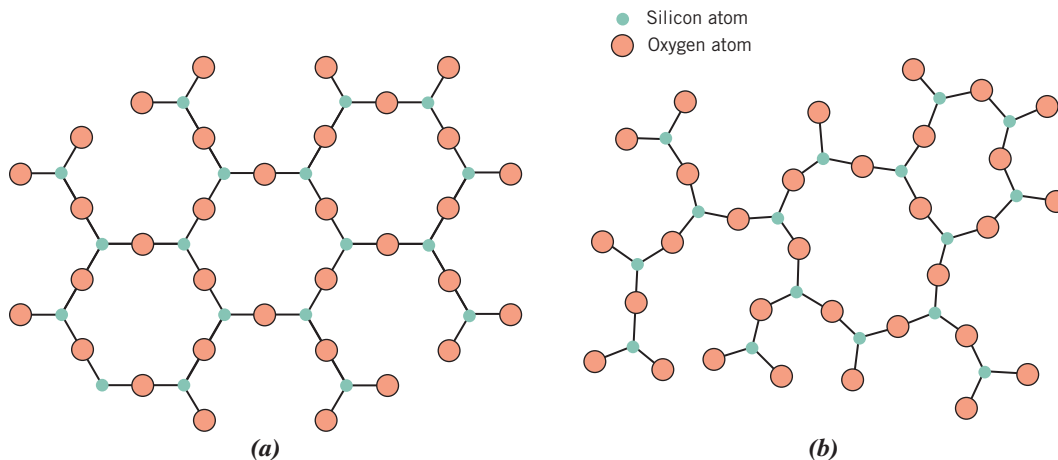


Figure 3.24 Two-dimensional schemes of the structure of (a) crystalline silicon dioxide and (b) noncrystalline silicon dioxide.

difficulty. Furthermore, rapidly cooling through the freezing temperature favors the formation of a noncrystalline solid, because little time is allowed for the ordering process.

Metals normally form crystalline solids, but some ceramic materials are crystalline, whereas others—the inorganic glasses—are amorphous. Polymers may be completely noncrystalline or semicrystalline consisting of varying degrees of crystallinity. More about the structure and properties of amorphous ceramics and polymers is contained in Chapters 12 and 14.



Concept Check 3.4 Do noncrystalline materials display the phenomenon of allotropy (or polymorphism)? Why or why not?

[The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

SUMMARY

Fundamental Concepts

- Atoms in crystalline solids are positioned in orderly and repeated patterns that are in contrast to the random and disordered atomic distribution found in noncrystalline or amorphous materials.

Unit Cells

- Crystal structures are specified in terms of parallelepiped unit cells, which are characterized by geometry and atom positions within.

Metallic Crystal Structures

- Most common metals exist in at least one of three relatively simple crystal structures: Face-centered cubic (FCC), which has a cubic unit cell (Figure 3.1). Body-centered cubic (BCC), which also has a cubic unit cell (Figure 3.2). Hexagonal close-packed, which has a unit cell of hexagonal symmetry, [Figure 3.4(a)].
- Two features of a crystal structure are
 Coordination number—the number of nearest-neighbor atoms, and
 Atomic packing factor—the fraction of solid sphere volume in the unit cell.

- Density Computations**

 - The theoretical density of a metal (ρ) is a function of the number of equivalent atoms per unit cell, the atomic weight, unit cell volume, and Avogadro's number (Equation 3.8).
- Polymorphism and Allotropy**

 - *Polymorphism* is when a specific material can have more than one crystal structure. *Allotropy* is polymorphism for elemental solids.
- Crystal Systems**

 - The concept of a crystal system is used to classify crystal structures on the basis of unit cell geometry—that is, unit cell edge lengths and interaxial angles. There are seven crystal systems: cubic, tetragonal, hexagonal, orthorhombic, rhombohedral (trigonal), monoclinic, and triclinic.
- Point Coordinates**

 - Crystallographic points, directions, and planes are specified in terms of indexing schemes. The basis for the determination of each index is a coordinate axis system defined by the unit cell for the particular crystal structure.
- Crystallographic Directions**

The location of a point within a unit cell is specified using coordinates that are fractional multiples of the cell edge lengths (Equations 3.9a–3.9c).
- Crystallographic Planes**

Directional indices are computed in terms of differences between vector head and tail coordinates (Equations 3.10a–3.10c).

Planar (or Miller) indices are determined from the reciprocals of axial intercepts (Equations 3.13a–3.13c).

 - For hexagonal unit cells, a four-index scheme for both directions and planes is found to be more convenient. Directions may be determined using Equations 3.11a–3.11d and 3.12a–3.12c.
- Linear and Planar Densities**

 - Crystallographic directional and planar equivalencies are related to atomic linear and planar densities, respectively.
 - For a given crystal structure, planes having identical atomic packing yet different Miller indices belong to the same family.
- Close-Packed Crystal Structures**

 - Both FCC and HCP crystal structures may be generated by the stacking of close-packed planes of atoms on top of one another. With this scheme *A*, *B*, and *C* denote possible atom positions on a close-packed plane.

The stacking sequence for HCP is *ABABAB*. . . .

The stacking sequence for FCC is *ABCABCABC*. . . .
- Single Crystals**

 - *Single crystals* are materials in which the atomic order extends uninterrupted over the entirety of the specimen; under some circumstances, single crystals may have flat faces and regular geometric shapes.
- Polycrystalline Materials**

 - The vast majority of crystalline solids, however, are *polycrystalline*, being composed of many small crystals or grains having different crystallographic orientations.
 - A *grain boundary* is the boundary region separating two grains where there is some atomic mismatch.
- Anisotropy**

 - *Anisotropy* is the directionality dependence of properties. For isotropic materials, properties are independent of the direction of measurement.
- X-Ray Diffraction: Determination of Crystal Structures**

 - *X-ray diffractometry* is used for crystal structure and interplanar spacing determinations. A beam of x-rays directed on a crystalline material may experience diffraction (constructive interference) as a result of its interaction with a series of parallel atomic planes.
 - Bragg's law specifies the condition for diffraction of x-rays—Equation 3.20.
- Noncrystalline Solids**

 - Noncrystalline solid materials lack a systematic and regular arrangement of atoms or ions over relatively large distances (on an atomic scale). Sometimes the term *amorphous* is also used to describe these materials.

Equation Summary

Equation Number	Equation	Solving For
3.1	$a = 2R\sqrt{2}$	Unit cell edge length, FCC
3.3	$APF = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} = \frac{V_S}{V_C}$	Atomic packing factor
3.4	$a = \frac{4R}{\sqrt{3}}$	Unit cell edge length, BCC
3.8	$\rho = \frac{nA}{V_C N_A}$	Theoretical density of a metal
3.9a	$q = \frac{a}{P_x}$	Point index referenced to x axis
3.10a	$u = n\left(\frac{x_2 - x_1}{a}\right)$	Direction index referenced to x axis
3.11a	$u = \frac{1}{3}(2U - V)$	Direction index conversion to hexagonal
3.12a	$U = n\left(\frac{a_1'' - a_1'}{a}\right)$	Hexagonal direction index referenced to a_1 axis (three-axis scheme)
3.13a	$h = \frac{na}{A}$	Planar (Miller) index referenced to x axis
3.15	$LD = \frac{\text{number of atoms centered on direction vector}}{\text{length of direction vector}}$	Linear density
3.17	$PD = \frac{\text{number of atoms centered on a plane}}{\text{area of plane}}$	Planar density
3.20	$n\lambda = 2d_{hkl}\sin\theta$	Bragg's law; wavelength–interplanar spacing–angle of diffracted beam
3.21	$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$	Interplanar spacing for crystals having cubic symmetry

List of Symbols

Symbol	Meaning
a	Unit cell edge length for cubic structures; unit cell x -axial length
a_1'	Vector tail coordinate, hexagonal
a_1''	Vector head coordinate, hexagonal
A	Atomic weight
A	Planar intercept on x axis
d_{hkl}	Interplanar spacing for crystallographic planes having indices h , k , and l
n	Order of reflection for x -ray diffraction
n	Number of atoms associated with a unit cell

(continued)

<i>Symbol</i>	<i>Meaning</i>
n	Normalization factor—reduction of directional/planar indices to integers
N_A	Avogadro's number (6.022×10^{23} atoms/mol)
P_x	Lattice position coordinate
R	Atomic radius
V_C	Unit cell volume
x_1	Vector tail coordinate
x_2	Vector head coordinate
λ	X-ray wavelength
ρ	Density; theoretical density

Important Terms and Concepts

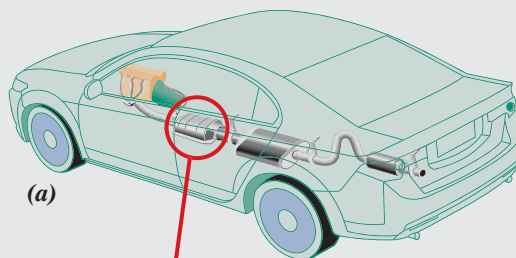
allotropy	crystal structure	lattice
amorphous	crystal system	lattice parameters
anisotropy	diffraction	Miller indices
atomic packing factor (APF)	face-centered cubic (FCC)	noncrystalline
body-centered cubic (BCC)	grain	polycrystalline
Bragg's law	grain boundary	polymorphism
coordination number	hexagonal close-packed (HCP)	single crystal
crystalline	isotropic	unit cell

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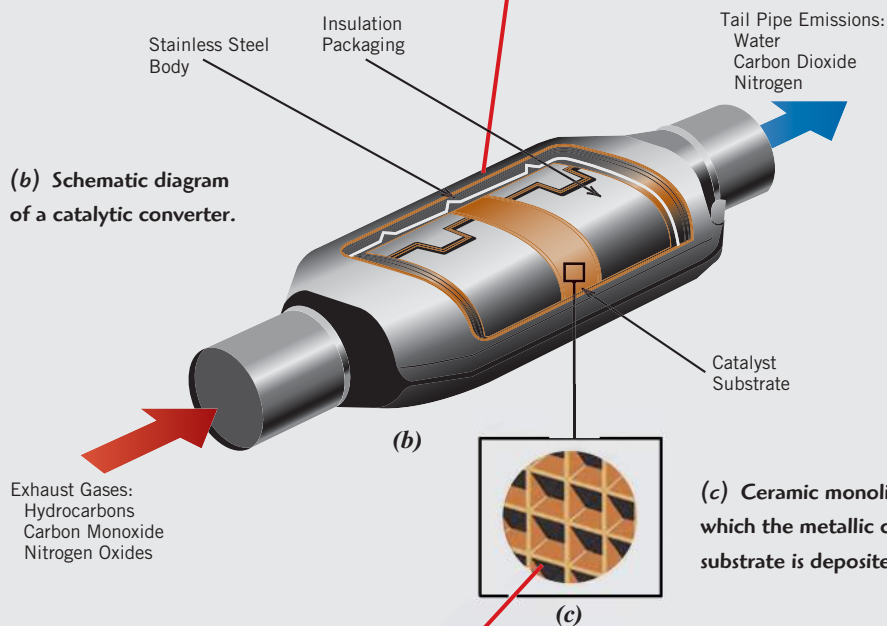
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Chapter 4 Imperfections in Solids

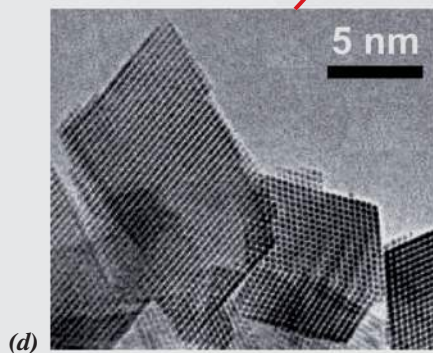
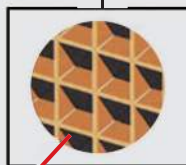
(a) Schematic diagram showing the location of the catalytic converter in an automobile's exhaust system.



(b) Schematic diagram of a catalytic converter.



(c) Ceramic monolith on which the metallic catalyst substrate is deposited.



(d) High-resolution transmission electron micrograph that shows surface defects on single crystals of one material that is used in catalytic converters.

Atomic defects are responsible for reductions of gas pollutant emissions from today's automobile engines. A catalytic converter is the pollutant-reducing device that is located in the automobile's exhaust system. Molecules of pollutant gases become attached to surface defects of crystalline metallic materials found in the catalytic converter. While attached to these sites, the molecules experience chemical reactions that convert them into other, non-polluting or less-polluting substances. The Materials of Importance box in Section 4.6 contains a detailed description of this process.

[Figure (d) from W. J. Stark, L. Mädler, M. Maciejewski, S. E. Pratsinis, and A. Baiker, "Flame-Synthesis of Nanocrystalline Ceria/Zirconia: Effect of Carrier Liquid," *Chem. Comm.*, 588–589 (2003). Reproduced by permission of The Royal Society of Chemistry.]

WHY STUDY *Imperfections in Solids*?

The properties of some materials are profoundly influenced by the presence of imperfections. Consequently, it is important to have a knowledge about the types of imperfections that exist and the roles they play in affecting the behavior of materials. For example, the mechanical properties of pure metals experience significant alterations when the metals are alloyed (i.e., when impurity atoms are added)—for example, brass (70% copper–30% zinc)

is much harder and stronger than pure copper (Section 7.9).

Also, integrated-circuit microelectronic devices found in our computers, calculators, and home appliances function because of highly controlled concentrations of specific impurities that are incorporated into small, localized regions of semiconducting materials (Sections 18.11 and 18.15).

Learning Objectives

After studying this chapter, you should be able to do the following:

1. Describe both vacancy and self-interstitial crystalline defects.
2. Calculate the equilibrium number of vacancies in a material at some specified temperature, given the relevant constants.
3. Name the two types of solid solutions and provide a brief written definition and/or schematic sketch of each.
4. Given the masses and atomic weights of two or more elements in a metal alloy, calculate the weight percent and atom percent for each element.
5. For each of edge, screw, and mixed dislocations:
 - (a) describe and make a drawing of the dislocation,
 - (b) note the location of the dislocation line, and
 - (c) indicate the direction along which the dislocation line extends.
6. Describe the atomic structure within the vicinity of
 - (a) a grain boundary and (b) a twin boundary.

4.1 INTRODUCTION

imperfection

Thus far it has been tacitly assumed that perfect order exists throughout crystalline materials on an atomic scale. However, such an idealized solid does not exist; all contain large numbers of various defects or **imperfections**. As a matter of fact, many of the properties of materials are profoundly sensitive to deviations from crystalline perfection; the influence is not always adverse, and often specific characteristics are deliberately fashioned by the introduction of controlled amounts or numbers of particular defects, as detailed in succeeding chapters.

point defect

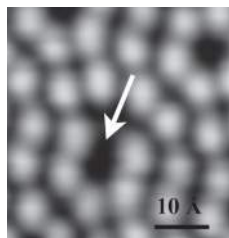
A *crystalline defect* refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. Classification of crystalline imperfections is frequently made according to the geometry or dimensionality of the defect. Several different imperfections are discussed in this chapter, including **point defects** (those associated with one or two atomic positions); linear (or one-dimensional) defects; and interfacial defects, or boundaries, which are two-dimensional. Impurities in solids are also discussed, because impurity atoms may exist as point defects. Finally, techniques for the microscopic examination of defects and the structure of materials are briefly described.

Point Defects

4.2 VACANCIES AND SELF-INTERSTITIALS

vacancy

The simplest of the point defects is a **vacancy**, or vacant lattice site, one normally occupied but from which an atom is missing (Figure 4.1). All crystalline solids contain vacancies, and, in fact, it is not possible to create such a material that is free of these



Scanning probe micrograph that shows a vacancy on a (111)-type surface plane for silicon. Approximately 7,000,000 $\times$. (Micrograph courtesy of D. Huang, Stanford University.)

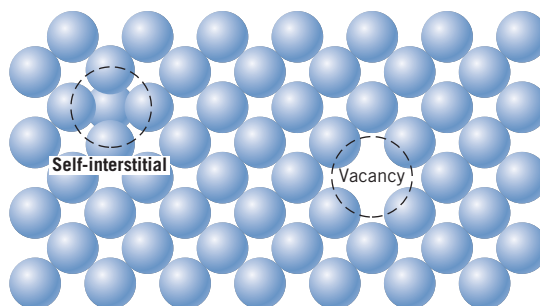


Figure 4.1 Two-dimensional representations of a vacancy and a self-interstitial.

(Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.)

defects. The necessity of the existence of vacancies is explained using principles of thermodynamics; in essence, the presence of vacancies increases the entropy (i.e., the randomness) of the crystal.

The equilibrium number of vacancies N_v for a given quantity of material (usually per meter cubed) depends on and increases with temperature according to

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right) \quad (4.1)$$

Temperature dependence of the equilibrium number of vacancies

Boltzmann's constant

In this expression, N is the total number of atomic sites (most commonly per cubic meter), Q_v is the energy required for the formation of a vacancy (J/mol or eV/atom), T is the absolute temperature in kelvins,¹ and k is the gas or **Boltzmann's constant**. The value of k is 1.38×10^{-23} J/atom·K, or 8.62×10^{-5} eV/atom·K, depending on the units of Q_v .² Thus, the number of vacancies increases exponentially with temperature—that is, as T in Equation 4.1 increases, so also does the term $\exp(-Q_v/kT)$. For most metals, the fraction of vacancies N_v/N just below the melting temperature is on the order of 10^{-4} —that is, one lattice site out of 10,000 will be empty. As ensuing discussions indicate, a number of other material parameters have an exponential dependence on temperature similar to that in Equation 4.1.

self-interstitial

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Tutorial Video:
Computation of
the Equilibrium
Number of
Vacancies

A **self-interstitial** is an atom from the crystal that is crowded into an *interstitial site*—a small void space that under ordinary circumstances is not occupied. This kind of defect is also represented in Figure 4.1. In metals, a self-interstitial introduces relatively large distortions in the surrounding lattice because the atom is substantially larger than the interstitial position in which it is situated. Consequently, the formation of this defect is not highly probable, and it exists in very small concentrations that are significantly lower than for vacancies.

¹Absolute temperature in kelvins (K) is equal to $^{\circ}\text{C} + 273$.

²Boltzmann's constant per mole of atoms becomes the gas constant R ; in such a case, $R = 8.31$ J/mol·K.

EXAMPLE PROBLEM 4.1**Number-of-Vacancies Computation at a Specified Temperature**

Calculate the equilibrium number of vacancies per cubic meter for copper at 1000°C. The energy for vacancy formation is 0.9 eV/atom; the atomic weight and density (at 1000°C) for copper are 63.5 g/mol and 8.4 g/cm³, respectively.

Solution

This problem may be solved by using Equation 4.1; it is first necessary, however, to determine the value of N_{Cu} —the number of atomic sites per cubic meter for copper, from its atomic weight A_{Cu} , its density ρ , and Avogadro's number N_A , according to

$$\begin{aligned} N_{\text{Cu}} &= \frac{N_A \rho}{A_{\text{Cu}}} \\ &= \frac{(6.022 \times 10^{23} \text{ atoms/mol})(8.4 \text{ g/cm}^3)(10^6 \text{ cm}^3/\text{m}^3)}{63.5 \text{ g/mol}} \\ &= 8.0 \times 10^{28} \text{ atoms/m}^3 \end{aligned} \quad (4.2)$$

Thus, the number of vacancies at 1000°C (1273 K) is equal to

$$\begin{aligned} N_v &= N \exp\left(-\frac{Q_v}{kT}\right) \\ &= (8.0 \times 10^{28} \text{ atoms/m}^3) \exp\left[-\frac{(0.9 \text{ eV})}{(8.62 \times 10^{-5} \text{ eV/K})(1273 \text{ K})}\right] \\ &= 2.2 \times 10^{25} \text{ vacancies/m}^3 \end{aligned}$$

Number of atoms per unit volume for a metal

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Tutorial Video

4.3 IMPURITIES IN SOLIDS

alloy

A pure metal consisting of only one type of atom just isn't possible; impurity or foreign atoms are always present, and some exist as crystalline point defects. In fact, even with relatively sophisticated techniques, it is difficult to refine metals to a purity in excess of 99.9999%. At this level, on the order of 10^{22} to 10^{23} impurity atoms are present in 1 m³ of material. Most familiar metals are not highly pure; rather, they are **alloys**, in which impurity atoms have been added intentionally to impart specific characteristics to the material. Ordinarily, alloying is used in metals to improve mechanical strength and corrosion resistance. For example, sterling silver is a 92.5% silver–7.5% copper alloy. In normal ambient environments, pure silver is highly corrosion resistant, but also very soft. Alloying with copper significantly enhances the mechanical strength without reducing the corrosion resistance appreciably.

solid solution

The addition of impurity atoms to a metal results in the formation of a **solid solution** and/or a new *second phase*, depending on the kinds of impurity, their concentrations, and the temperature of the alloy. The present discussion is concerned with the notion of a solid solution; treatment of the formation of a new phase is deferred to Chapter 9.

solute, solvent

Several terms relating to impurities and solid solutions deserve mention. With regard to alloys, **solute** and **solvent** are terms that are commonly employed. *Solvent* is the element or compound that is present in the greatest amount; on occasion, solvent atoms are also called *host atoms*. *Solute* is used to denote an element or compound present in a minor concentration.

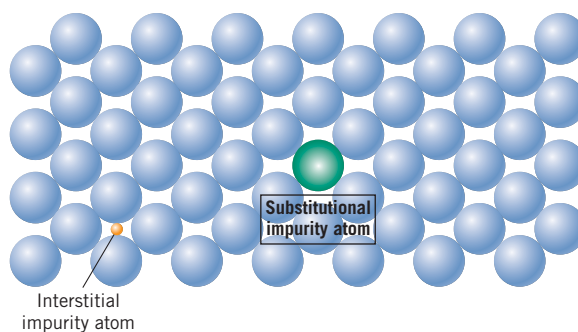


Figure 4.2 Two-dimensional schematic representations of substitutional and interstitial impurity atoms. (Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.)

Solid Solutions

A solid solution forms when, as the solute atoms are added to the host material, the crystal structure is maintained and no new structures are formed. Perhaps it is useful to draw an analogy with a liquid solution. If two liquids that are soluble in each other (such as water and alcohol) are combined, a liquid solution is produced as the molecules intermix, and its composition is homogeneous throughout. A solid solution is also compositionally homogeneous; the impurity atoms are randomly and uniformly dispersed within the solid.

Impurity point defects are found in solid solutions, of which there are two types: **substitutional** and **interstitial**. For the substitutional type, solute or impurity atoms replace or substitute for the host atoms (Figure 4.2). Several features of the solute and solvent atoms determine the degree to which the former dissolves in the latter. These are expressed as four *Hume–Rothery rules*, as follows:

1. **Atomic size factor.** Appreciable quantities of a solute may be accommodated in this type of solid solution only when the difference in atomic radii between the two atom types is less than about $\pm 15\%$. Otherwise, the solute atoms create substantial lattice distortions and a new phase forms.
2. **Crystal structure.** For appreciable solid solubility, the crystal structures for metals of both atom types must be the same.
3. **Electronegativity factor.** The more electropositive one element and the more electronegative the other, the greater the likelihood that they will form an intermetallic compound instead of a substitutional solid solution.
4. **Valences.** Other factors being equal, a metal has more of a tendency to dissolve another metal of higher valency than to dissolve one of a lower valency.

An example of a substitutional solid solution is found for copper and nickel. These two elements are completely soluble in one another at all proportions. With regard to the aforementioned rules that govern degree of solubility, the atomic radii for copper and nickel are 0.128 and 0.125 nm, respectively; both have the FCC crystal structure; and their electronegativities are 1.9 and 1.8 (Figure 2.9). Finally, the most common valences are +1 for copper (although it sometimes can be +2) and +2 for nickel.

For interstitial solid solutions, impurity atoms fill the voids or interstices among the host atoms (see Figure 4.2). For both FCC and BCC crystal structures, there are two types of interstitial sites—*tetrahedral* and *octahedral*; these are distinguished by the number of nearest neighbor host atoms—that is, the coordination number. Tetrahedral sites

substitutional solid solution

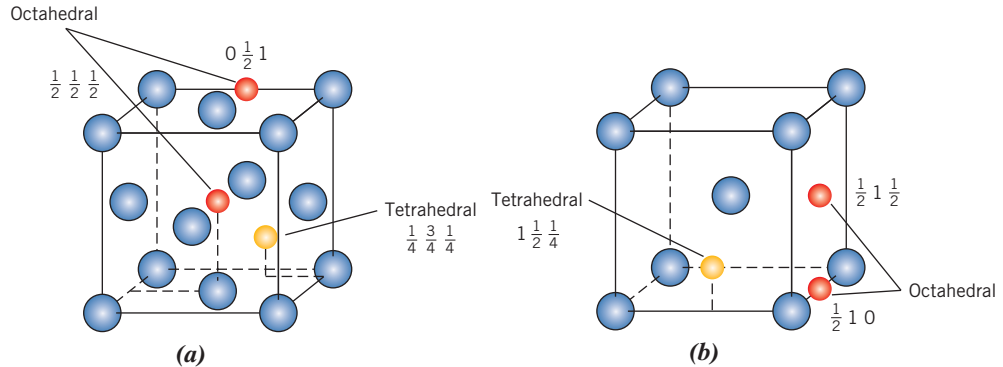
interstitial solid solution

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Tutorial Video:
Defects
What Are
the Differences
between Interstitial
and Substitutional
Solid Solutions?

Figure 4.3

Locations of tetrahedral and octahedral interstitial sites within (a) FCC and (b) BCC unit cells.



have a coordination number of 4; straight lines drawn from the centers of the surrounding host atoms form a four-sided tetrahedron. However, for octahedral sites the coordination number is 6; an octahedron is produced by joining these six sphere centers.³ For FCC, there are two types of octahedral sites with representative point coordinates of $0, \frac{1}{2}, \frac{1}{2}$ and $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$. Representative coordinates for a single tetrahedral site type are $\frac{1}{4}, \frac{3}{4}, \frac{1}{4}$.⁴ Locations of these sites within the FCC unit cell are noted in Figure 4.3a. Likewise, for BCC, there are two octahedral and one tetrahedral types. Representative coordinates for the octahedral sites are as follows: $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ and $\frac{1}{2}, \frac{1}{2}, 0$; for BCC tetrahedral, $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ is a representative coordinate. Figure 4.3b shows the positions of these sites within a BCC unit cell.⁴

Metallic materials have relatively high atomic packing factors, which means that these interstitial positions are relatively small. Consequently, the atomic diameter of an interstitial impurity must be substantially smaller than that of the host atoms. Normally, the maximum allowable concentration of interstitial impurity atoms is low (less than 10%). Even very small impurity atoms are ordinarily larger than the interstitial sites, and as a consequence, they introduce some lattice strains on the adjacent host atoms. Problems 4.8 and 4.9 call for determination of the radii of impurity atoms r (in terms of R , the host atom radius) that just fit into tetrahedral and octahedral interstitial positions of both BCC and FCC without introducing any lattice strains.

Carbon forms an interstitial solid solution when added to iron; the maximum concentration of carbon is about 2%. The atomic radius of the carbon atom is much less than that of iron: 0.071 nm versus 0.124 nm.

Solid solutions are also possible for ceramic materials, as discussed in Section 12.5.

EXAMPLE PROBLEM 4.2

Computation of Radius of BCC Interstitial Site

Compute the radius r of an impurity atom that just fits into a BCC octahedral site in terms of the atomic radius R of the host atom (without introducing lattice strains).

Solution

As Figure 4.3b notes, for BCC, one octahedral interstitial site is situated at the center of a unit cell edge. In order for an interstitial atom to be positioned in this site without introducing lattice

³The geometries of these site types may be observed in Figure 12.7.

⁴Other octahedral and tetrahedral interstices are located at positions within the unit cell that are equivalent to these representative ones.

strains, the atom just touches the two adjacent host atoms, which are corner atoms of the unit cell. The drawing shows atoms on the (100) face of a BCC unit cell; the large circles represent the host atoms—the small circle represents an interstitial atom that is positioned in an octahedral site on the cube edge.

On this drawing is noted the unit cell edge length—the distance between the centers of the corner atoms—which, from Equation 3.4, is equal to

$$\text{Unit cell edge length} = \frac{4R}{\sqrt{3}}$$

Also shown is that the unit cell edge length is equal to two times the sum of host atomic radius $2R$ plus twice the radius of the interstitial atom $2r$; i.e.,

$$\text{Unit cell edge length} = 2R + 2r$$

Now, equating these two unit cell edge length expressions, we get

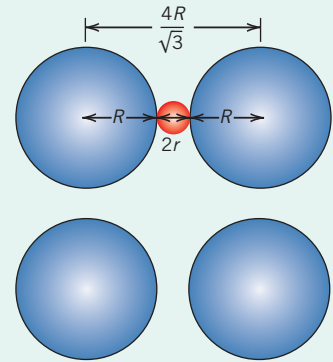
$$2R + 2r = \frac{4R}{\sqrt{3}}$$

and solving for r in terms of R

$$2r = \frac{4R}{\sqrt{3}} - 2R = \left(\frac{2}{\sqrt{3}} - 1 \right) (2R)$$

or

$$r = \left(\frac{2}{\sqrt{3}} - 1 \right) R = 0.155R$$



Concept Check 4.1 Is it possible for three or more elements to form a solid solution? Explain your answer.

[The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]



Concept Check 4.2 Explain why complete solid solubility may occur for substitutional solid solutions but not for interstitial solid solutions.

[The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

4.4 SPECIFICATION OF COMPOSITION

composition

It is often necessary to express the **composition** (or *concentration*)⁵ of an alloy in terms of its constituent elements. The two most common ways to specify composition are weight (or mass) percent and atom percent. The basis for **weight percent** (wt%) is the weight of a particular element relative to the total alloy weight. For an alloy that

weight percent

⁵The terms *composition* and *concentration* will be assumed to have the same meaning in this book (i.e., the relative content of a specific element or constituent in an alloy) and will be used interchangeably.

contains two hypothetical atoms denoted by 1 and 2, the concentration of 1 in wt%, C_1 , is defined as

Computation of
weight percent (for a
two-element alloy)

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100 \quad (4.3a)$$

where m_1 and m_2 represent the weight (or mass) of elements 1 and 2, respectively. The concentration of 2 is computed in an analogous manner.⁶

atom percent

The basis for **atom percent** (at%) calculations is the number of moles of an element in relation to the total moles of the elements in the alloy. The number of moles in some specified mass of a hypothetical element 1, n_{m1} , may be computed as follows:

$$n_{m1} = \frac{m'_1}{A_1} \quad (4.4)$$

Here, m'_1 and A_1 denote the mass (in grams) and atomic weight, respectively, for element 1.

Concentration in terms of atom percent of element 1 in an alloy containing element 1 and element 2 atoms, C'_1 is defined by⁷

Computation of
atom percent (for a
two-element alloy)

$$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100 \quad (4.5a)$$

In like manner, the atom percent of element 2 is determined.⁸

Atom percent computations also can be carried out on the basis of the number of atoms instead of moles, because one mole of all substances contains the same number of atoms.

Composition Conversions

Sometimes it is necessary to convert from one composition scheme to another—for example, from weight percent to atom percent. We next present equations for making these conversions in terms of the two hypothetical elements 1 and 2. Using the convention of the previous section (i.e., weight percents denoted by C_1 and C_2 , atom percents by C'_1 and C'_2 , and atomic weights as A_1 and A_2), we express these conversion equations as follows:

Conversion of
weight percent to
atom percent (for a
two-element alloy)

$$C'_1 = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100 \quad (4.6a)$$

$$C'_2 = \frac{C_2 A_1}{C_1 A_2 + C_2 A_1} \times 100 \quad (4.6b)$$

⁶When an alloy contains more than two (say n) elements, Equation (4.3a) takes the form

$$C_1 = \frac{m_1}{m_1 + m_2 + m_3 + \cdots + m_n} \times 100 \quad (4.3b)$$

⁷In order to avoid confusion in notations and symbols being used in this section, we should point out that the prime (as in C'_1 and m'_1) is used to designate both composition in atom percent and mass of material in grams.

⁸When an alloy contains more than two (say n) elements, Equation (4.5a) takes the form

$$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2} + n_{m3} + \cdots + n_{mn}} \times 100 \quad (4.5b)$$

Conversion of
atom percent to
weight percent (for
a two-element alloy)

$$C_1 = \frac{C'_1 A_1}{C'_1 A_1 + C'_2 A_2} \times 100 \quad (4.7a)$$

$$C_2 = \frac{C'_2 A_2}{C'_1 A_1 + C'_2 A_2} \times 100 \quad (4.7b)$$

Because we are considering only two elements, computations involving the preceding equations are simplified when it is realized that

$$C_1 + C_2 = 100 \quad (4.8a)$$

$$C'_1 + C'_2 = 100 \quad (4.8b)$$

In addition, it sometimes becomes necessary to convert concentration from weight percent to mass of one component per unit volume of material (i.e., from units of wt% to kg/m³); this latter composition scheme is often used in diffusion computations (Section 5.3). Concentrations in terms of this basis are denoted using a double prime (i.e., C''_1 and C''_2), and the relevant equations are as follows:

$$C''_1 = \left(\frac{C_1}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3 \quad (4.9a)$$

$$C''_2 = \left(\frac{C_2}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3 \quad (4.9b)$$

Conversion of weight
percent to mass per
unit volume (for a
two-element alloy)

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Tutorial Video:

Weight Percent
and Atom Percent
Calculations

For density ρ in units of g/cm³, these expressions yield C''_1 and C''_2 in kg/m³.

Furthermore, on occasion we desire to determine the density and atomic weight of a binary alloy, given the composition in terms of either weight percent or atom percent. If we represent alloy density and atomic weight by ρ_{ave} and A_{ave} , respectively, then

$$\rho_{\text{ave}} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \quad (4.10a)$$

$$\rho_{\text{ave}} = \frac{\frac{C'_1 A_1}{\rho_1} + \frac{C'_2 A_2}{\rho_2}}{\frac{C'_1 A_1}{\rho_1} + \frac{C'_2 A_2}{\rho_2}} \quad (4.10b)$$

$$A_{\text{ave}} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}} \quad (4.11a)$$

$$A_{\text{ave}} = \frac{\frac{C'_1 A_1}{100} + \frac{C'_2 A_2}{100}}{\frac{C'_1 A_1}{100} + \frac{C'_2 A_2}{100}} \quad (4.11b)$$

Computation of
density (for a two-
element metal alloy)

Computation of
atomic weight (for a
two-element metal
alloy)

It should be noted that Equations 4.9 and 4.11 are not always exact. In their derivations, it is assumed that total alloy volume is exactly equal to the sum of the volumes of the individual elements. This normally is not the case for most alloys; however, it is a

reasonably valid assumption and does not lead to significant errors for dilute solutions and over composition ranges where solid solutions exist.

EXAMPLE PROBLEM 4.3

Derivation of Composition-Conversion Equation

Derive Equation 4.6a.

Solution

To simplify this derivation, we assume that masses are expressed in units of grams and denoted with a prime (e.g., m'_1). Furthermore, the total alloy mass (in grams) M' is

$$M' = m'_1 + m'_2 \quad (4.12)$$

Using the definition of C'_1 (Equation 4.5a) and incorporating the expression for n_{m1} , Equation 4.4, and the analogous expression for n_{m2} yields

$$\begin{aligned} C'_1 &= \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100 \\ &= \frac{\frac{m'_1}{A_1}}{\frac{m'_1}{A_1} + \frac{m'_2}{A_2}} \times 100 \end{aligned} \quad (4.13)$$

Rearrangement of the mass-in-grams equivalent of Equation 4.3a leads to

$$m'_1 = \frac{C_1 M'}{100} \quad (4.14)$$

Substitution of this expression and its m'_2 equivalent into Equation 4.13 gives

$$C'_1 = \frac{\frac{C_1 M'}{100 A_1}}{\frac{C_1 M'}{100 A_1} + \frac{C_2 M'}{100 A_2}} \times 100 \quad (4.15)$$

Upon simplification, we have

$$C'_1 = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100$$

which is identical to Equation 4.6a.

EXAMPLE PROBLEM 4.4

Composition Conversion—From Weight Percent to Atom Percent

Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminum and 3 wt% copper.

Solution

If we denote the respective weight percent compositions as $C_{Al} = 97$ and $C_{Cu} = 3$, substitution into Equations 4.6a and 4.6b yields

WileyPLUS**Tutorial Video:**

How to
Convert from
Atom Percent
to Weight
Percent

and

$$C'_{\text{Al}} = \frac{C_{\text{Al}}A_{\text{Cu}}}{C_{\text{Al}}A_{\text{Cu}} + C_{\text{Cu}}A_{\text{Al}}} \times 100$$

$$= \frac{(97)(63.55 \text{ g/mol})}{(97)(63.55 \text{ g/mol}) + (3)(26.98 \text{ g/mol})} \times 100$$

$$= 98.7 \text{ at\%}$$

$$C'_{\text{Cu}} = \frac{C_{\text{Cu}}A_{\text{Al}}}{C_{\text{Cu}}A_{\text{Al}} + C_{\text{Al}}A_{\text{Cu}}} \times 100$$

$$= \frac{(3)(26.98 \text{ g/mol})}{(3)(26.98 \text{ g/mol}) + (97)(63.55 \text{ g/mol})} \times 100$$

$$= 1.30 \text{ at\%}$$

Miscellaneous Imperfections

4.5 DISLOCATIONS—LINEAR DEFECTS

edge dislocation
dislocation line



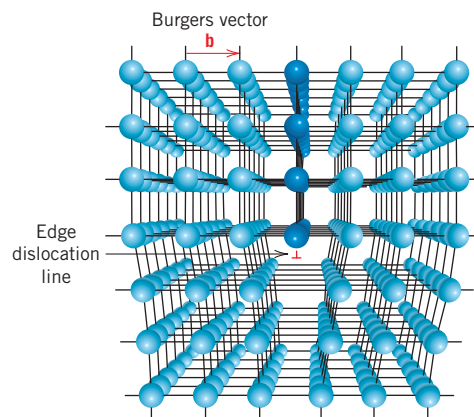
WileyPLUS: VMSE
Edge

screw dislocation

A *dislocation* is a linear or one-dimensional defect around which some of the atoms are misaligned. One type of dislocation is represented in Figure 4.4: an extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal. This is termed an **edge dislocation**; it is a linear defect that centers on the line that is defined along the end of the extra half-plane of atoms. This is sometimes termed the **dislocation line**, which, for the edge dislocation in Figure 4.4, is perpendicular to the plane of the page. Within the region around the dislocation line there is some localized lattice distortion. The atoms above the dislocation line in Figure 4.4 are squeezed together, and those below are pulled apart; this is reflected in the slight curvature for the vertical planes of atoms as they bend around this extra half-plane. The magnitude of this distortion decreases with distance away from the dislocation line; at positions far removed, the crystal lattice is virtually perfect. Sometimes the edge dislocation in Figure 4.4 is represented by the symbol $\perp$, which also indicates the position of the dislocation line. An edge dislocation may also be formed by an extra half-plane of atoms that is included in the bottom portion of the crystal; its designation is a T.

Another type of dislocation, called a **screw dislocation**, may be thought of as being formed by a shear stress that is applied to produce the distortion shown in Figure 4.5a: The

Figure 4.4 The atom positions around an edge dislocation; extra half-plane of atoms shown in perspective.





upper front region of the crystal is shifted one atomic distance to the right relative to the bottom portion. The atomic distortion associated with a screw dislocation is also linear and along a dislocation line, line AB in Figure 4.5b. The screw dislocation derives its name from the spiral or helical path or ramp that is traced around the dislocation line by the atomic planes of atoms. Sometimes the symbol ζ is used to designate a screw dislocation.

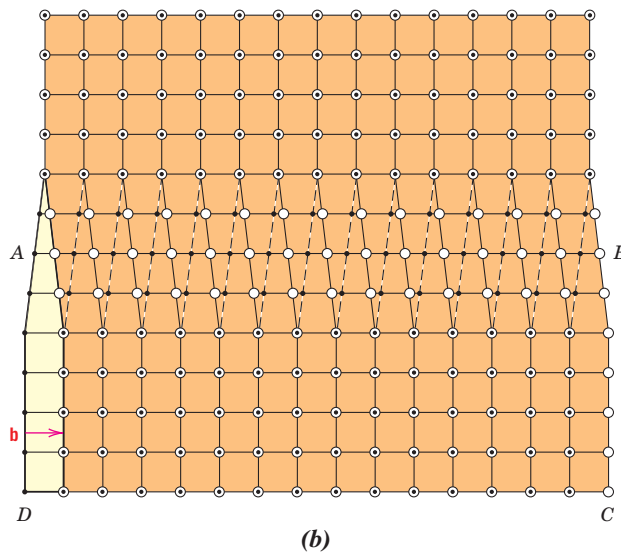
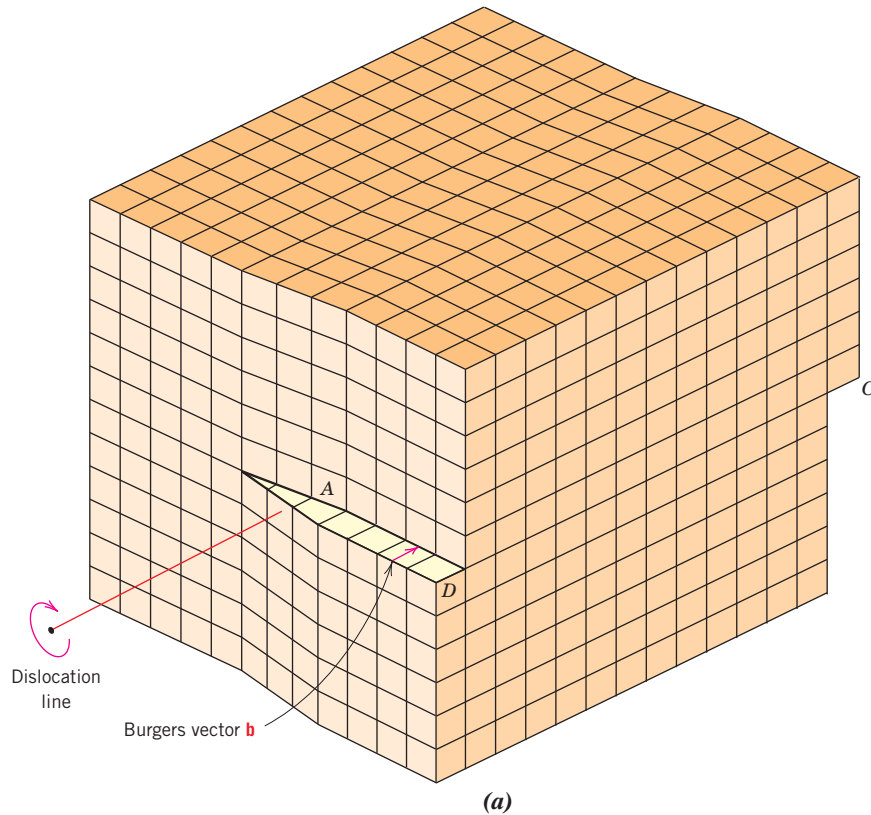


Figure 4.5 (a) A screw dislocation within a crystal. (b) The screw dislocation in (a) as viewed from above. The dislocation line extends along line AB . Atom positions above the slip plane are designated by open circles, those below by solid circles.

[Figure (b) from W. T. Read, Jr., *Dislocations in Crystals*, McGraw-Hill Book Company, New York, NY, 1953.]

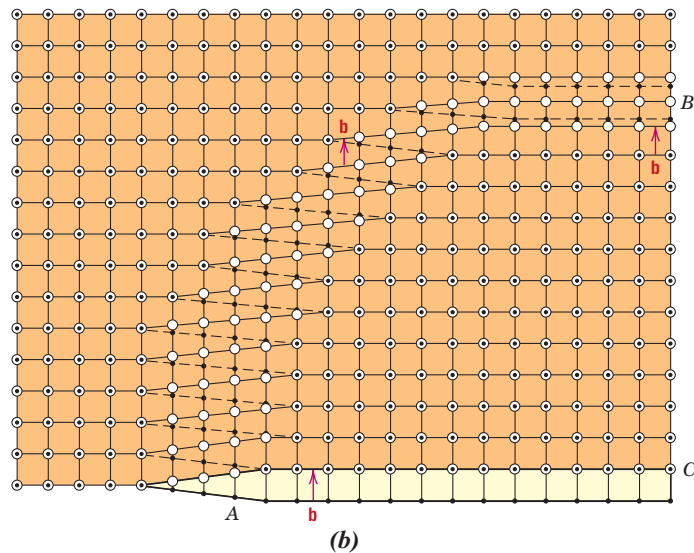
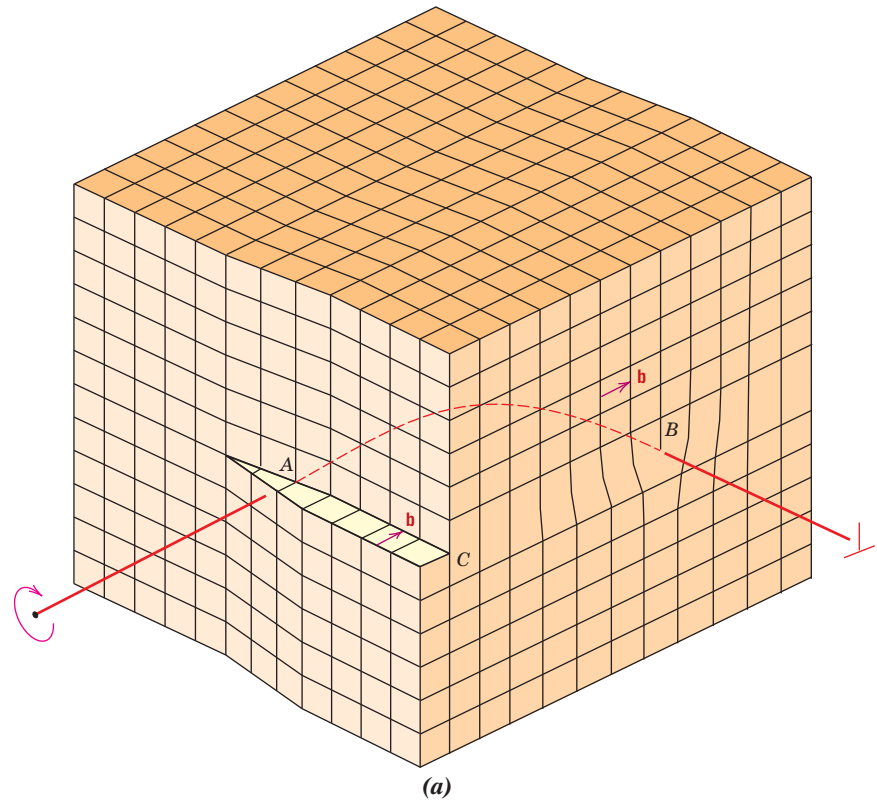
mixed dislocation

WileyPLUS: VMSE
Mixed

Most dislocations found in crystalline materials are probably neither pure edge nor pure screw but exhibit components of both types; these are termed **mixed dislocations**. All three dislocation types are represented schematically in Figure 4.6; the lattice distortion that is produced away from the two faces is mixed, having varying degrees of screw and edge character.

Figure 4.6 (a) Schematic representation of a dislocation that has edge, screw, and mixed character. (b) Top view, where open circles denote atom positions above the slip plane, and solid circles, atom positions below. At point *A*, the dislocation is pure screw, while at point *B*, it is pure edge. For regions in between where there is curvature in the dislocation line, the character is mixed edge and screw.

[Figure (b) from W. T. Read, Jr., *Dislocations in Crystals*, McGraw-Hill Book Company, New York, NY, 1953.]



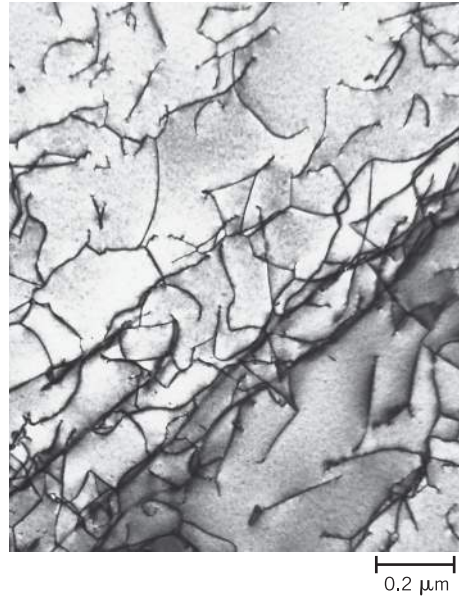


Figure 4.7 A transmission electron micrograph of a titanium alloy in which the dark lines are dislocations, 50,000 $\times$. (Courtesy of M. R. Plichta, Michigan Technological University.)

Burgers vector

The magnitude and direction of the lattice distortion associated with a dislocation are expressed in terms of a **Burgers vector**, denoted by **b**. Burgers vectors are indicated in Figures 4.4 and 4.5 for edge and screw dislocations, respectively. Furthermore, the nature of a dislocation (i.e., edge, screw, or mixed) is defined by the relative orientations of dislocation line and Burgers vector. For an edge, they are perpendicular (Figure 4.4), whereas for a screw, they are parallel (Figure 4.5); they are neither perpendicular nor parallel for a mixed dislocation. Also, even though a dislocation changes direction and nature within a crystal (e.g., from edge to mixed to screw), the Burgers vector is the same at all points along its line. For example, all positions of the curved dislocation in Figure 4.6 have the Burgers vector shown. For metallic materials, the Burgers vector for a dislocation points in a close-packed crystallographic direction and is of magnitude equal to the interatomic spacing.

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Tutorial Video: Defects

Screw and Edge
Dislocations

As we note in Section 7.2, the permanent deformation of most crystalline materials is by the motion of dislocations. In addition, the Burgers vector is an element of the theory that has been developed to explain this type of deformation.

Dislocations can be observed in crystalline materials using electron-microscopic techniques. In Figure 4.7, a high-magnification transmission electron micrograph, the dark lines are the dislocations.

Virtually all crystalline materials contain some dislocations that were introduced during solidification, during plastic deformation, and as a consequence of thermal stresses that result from rapid cooling. Dislocations are involved in the plastic deformation of crystalline materials, both metals and ceramics, as discussed in Chapters 7 and 12. They have also been observed in polymeric materials and are discussed in Section 14.13.

4.6 INTERFACIAL DEFECTS

Interfacial defects are boundaries that have two dimensions and normally separate regions of the materials that have different crystal structures and/or crystallographic orientations. These imperfections include external surfaces, grain boundaries, phase boundaries, twin boundaries, and stacking faults.

External Surfaces

One of the most obvious boundaries is the external surface, along which the crystal structure terminates. Surface atoms are not bonded to the maximum number of nearest neighbors and are therefore in a higher energy state than the atoms at interior positions. The bonds of these surface atoms that are not satisfied give rise to a surface energy, expressed in units of energy per unit area (J/m^2 or erg/cm^2). To reduce this energy, materials tend to minimize, if at all possible, the total surface area. For example, liquids assume a shape having a minimum area—the droplets become spherical. Of course, this is not possible with solids, which are mechanically rigid.

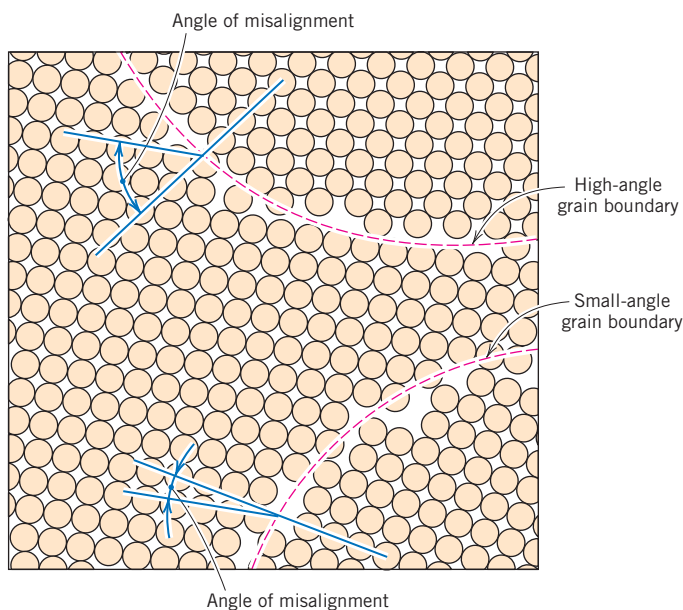
Grain Boundaries

Another interfacial defect, the grain boundary, was introduced in Section 3.14 as the boundary separating two small grains or crystals having different crystallographic orientations in polycrystalline materials. A grain boundary is represented schematically from an atomic perspective in Figure 4.8. Within the boundary region, which is probably just several atom distances wide, there is some atomic mismatch in a transition from the crystalline orientation of one grain to that of an adjacent one.

Various degrees of crystallographic misalignment between adjacent grains are possible (Figure 4.8). When this orientation mismatch is slight, on the order of a few degrees, then the term *small-* (or *low-*) *angle grain boundary* is used. These boundaries can be described in terms of dislocation arrays. One simple small-angle grain boundary is formed when edge dislocations are aligned in the manner of Figure 4.9. This type is called a *tilt boundary*; the angle of misorientation, θ , is also indicated in the figure. When the angle of misorientation is parallel to the boundary, a *twist boundary* results, which can be described by an array of screw dislocations.

The atoms are bonded less regularly along a grain boundary (e.g., bond angles are longer), and consequently there is an interfacial or grain boundary energy similar to the surface energy just described. The magnitude of this energy is a function of the degree of misorientation, being larger for high-angle boundaries. Grain boundaries are more chemically reactive than the grains themselves as a consequence of this boundary energy. Furthermore, impurity atoms often preferentially segregate along these boundaries because

Figure 4.8 Schematic diagram showing small- and high-angle grain boundaries and the adjacent atom positions.



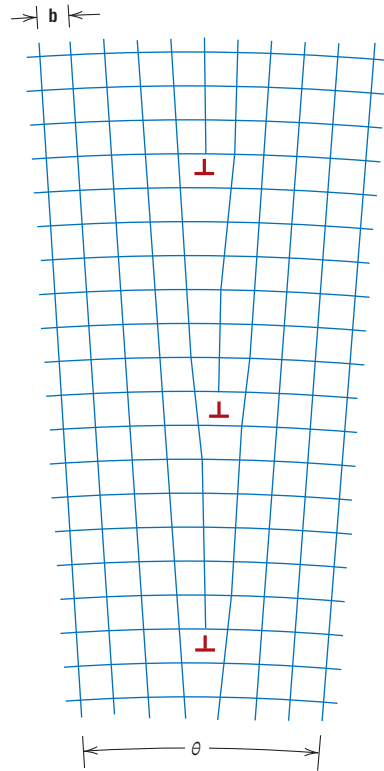


Figure 4.9 Demonstration of how a tilt boundary having an angle of misorientation θ results from an alignment of edge dislocations.

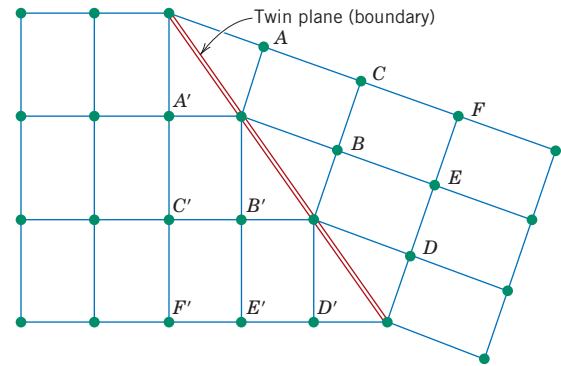


Figure 4.10 Schematic diagram showing a twin plane or boundary and the adjacent atom positions (colored circles). Atoms labeled with corresponding primed and unprimed letters (e.g., A and A') reside in mirror-imaged positions across the twin boundary.

of their higher energy state. The total interfacial energy is lower in large or coarse-grained materials than in fine-grained ones because there is less total boundary area in the former. Grains grow at elevated temperatures to reduce the total boundary energy, a phenomenon explained in Section 7.13.

In spite of this disordered arrangement of atoms and lack of regular bonding along grain boundaries, a polycrystalline material is still very strong; cohesive forces within and across the boundary are present. Furthermore, the density of a polycrystalline specimen is virtually identical to that of a single crystal of the same material.

Phase Boundaries

Phase boundaries exist in multiphase materials (Section 9.3), in which a different phase exists on each side of the boundary; In addition, each of the constituent phases has its own distinctive physical and/or chemical characteristics. As we shall see in subsequent chapters, phase boundaries play an important role in determining the mechanical characteristics of some multiphase metal alloys.

Twin Boundaries

A *twin boundary* is a special type of grain boundary across which there is a specific mirror lattice symmetry; that is, atoms on one side of the boundary are located in mirror-image positions to those of the atoms on the other side (Figure 4.10). The region

MATERIALS OF IMPORTANCE 4.1

Catalysts (and Surface Defects)

A *catalyst* is a substance that speeds up the rate of a chemical reaction without participating in the reaction itself (i.e., it is not consumed). One type of catalyst exists as a solid; reactant molecules in a gas or liquid phase are adsorbed⁹ onto the catalytic surface, at which point some type of interaction occurs that promotes an increase in their chemical reactivity rate.

Adsorption sites on a catalyst are normally surface defects associated with planes of atoms; an interatomic/intermolecular bond is formed between a defect site and an adsorbed molecular species. The several types of surface defects, represented schematically in Figure 4.11, include ledges, kinks, terraces, vacancies, and individual adatoms (i.e., atoms adsorbed on the surface).

One important use of catalysts is in catalytic converters on automobiles, which reduce the emission of exhaust gas pollutants such as carbon monoxide (CO), nitrogen oxides (NO_x , where x is variable), and unburned hydrocarbons. (See the chapter-opening diagrams and photograph for this chapter.) Air is introduced into the exhaust emissions from the automobile engine; this mixture of gases then passes over the catalyst, which on its surface adsorbs molecules of CO, NO_x , and O_2 . The NO_x dissociates into N and O atoms, whereas the O_2 dissociates into its atomic species. Pairs of nitrogen atoms combine to form N_2 molecules, and carbon monoxide is oxidized to form

carbon dioxide (CO_2). Furthermore, any unburned hydrocarbons are also oxidized to CO_2 and H_2O .

One of the materials used as a catalyst in this application is $(\text{Ce}_{0.5}\text{Zr}_{0.5})\text{O}_2$. Figure 4.12 is a high-resolution transmission electron micrograph that shows several single crystals of this material. Individual atoms are resolved in this micrograph as well as some of the defects presented in Figure 4.11. These surface defects act as adsorption sites for the atomic and molecular species noted in the previous paragraph. Consequently, dissociation, combination, and oxidation reactions involving these species are facilitated, such that the content of pollutant species (CO, NO_x , and unburned hydrocarbons) in the exhaust gas stream is reduced significantly.

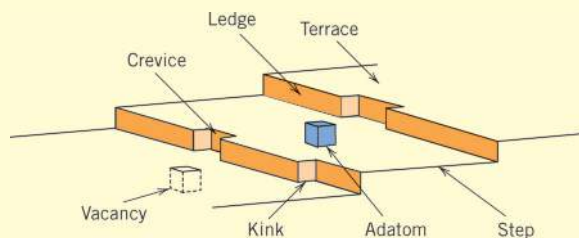


Figure 4.11 Schematic representations of surface defects that are potential adsorption sites for catalysis. Individual atom sites are represented as cubes.

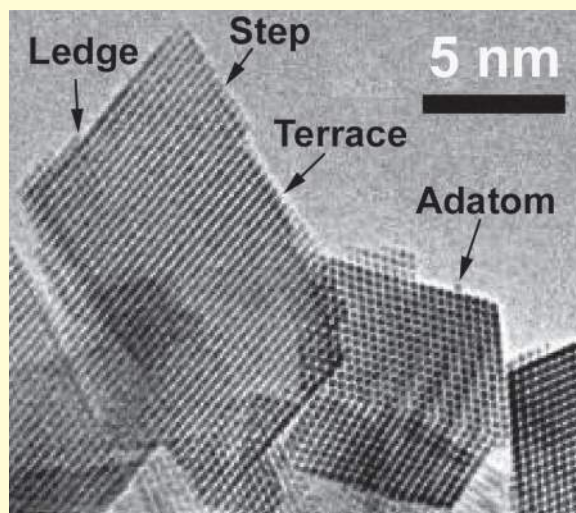


Figure 4.12 High-resolution transmission electron micrograph that shows single crystals of $(\text{Ce}_{0.5}\text{Zr}_{0.5})\text{O}_2$; this material is used in catalytic converters for automobiles. Surface defects represented schematically in Figure 4.11 are noted on the crystals.

[From W. J. Stark, L. Mädler, M. Maciejewski, S. E. Pratsinis, and A. Baiker, "Flame-Synthesis of Nanocrystalline Ceria/Zirconia: Effect of Carrier Liquid," *Chem. Comm.*, 588–589 (2003). Reproduced by permission of The Royal Society of Chemistry.]

⁹*Adsorption* is the adhesion of molecules of a gas or liquid to a solid surface. It should not be confused with *absorption*, which is the assimilation of molecules into a solid or liquid.

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Defects**Differences among
Point, Linear, and
Interfacial Defects

of material between these boundaries is appropriately termed a *twin*. Twins result from atomic displacements that are produced from applied mechanical shear forces (mechanical twins) and also during annealing heat treatments following deformation (annealing twins). Twinning occurs on a definite crystallographic plane and in a specific direction, both of which depend on the crystal structure. Annealing twins are typically found in metals that have the FCC crystal structure, whereas mechanical twins are observed in BCC and HCP metals. The role of mechanical twins in the deformation process is discussed in Section 7.7. Annealing twins may be observed in the photomicrograph of the polycrystalline brass specimen shown in Figure 4.14c. The twins correspond to those regions having relatively straight and parallel sides and a different visual contrast than the untwinned regions of the grains within which they reside. An explanation for the variety of textural contrasts in this photomicrograph is provided in Section 4.10.

Miscellaneous Interfacial Defects

Other possible interfacial defects include stacking faults and ferromagnetic domain walls. Stacking faults are found in FCC metals when there is an interruption in the *ABCABCABC* . . . stacking sequence of close-packed planes (Section 3.12). For ferromagnetic and ferrimagnetic materials, the boundary that separates regions having different directions of magnetization is termed a *domain wall*, which is discussed in Section 20.7.

Associated with each of the defects discussed in this section is an interfacial energy, the magnitude of which depends on boundary type, and which varies from material to material. Normally, the interfacial energy is greatest for external surfaces and least for domain walls.



Concept Check 4.3 The surface energy of a single crystal depends on crystallographic orientation. Does this surface energy increase or decrease with an increase in planar density? Why?

[The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

4.7 BULK OR VOLUME DEFECTS

Other defects exist in all solid materials that are much larger than those heretofore discussed. These include pores, cracks, foreign inclusions, and other phases. They are normally introduced during processing and fabrication steps. Some of these defects and their effects on the properties of materials are discussed in subsequent chapters.

4.8 ATOMIC VIBRATIONS**atomic vibration**

Every atom in a solid material is vibrating very rapidly about its lattice position within the crystal. In a sense, these **atomic vibrations** may be thought of as imperfections or defects. At any instant of time, not all atoms vibrate at the same frequency and amplitude or with the same energy. At a given temperature, there exists a distribution of energies for the constituent atoms about an average energy. Over time, the vibrational energy of any specific atom also varies in a random manner. With rising temperature, this average energy increases, and, in fact, the temperature of a solid is really just a measure of the average vibrational activity of atoms and molecules. At room temperature, a typical vibrational frequency is on the order of 10^{13} vibrations per second, whereas the amplitude is a few thousandths of a nanometer.

Many properties and processes in solids are manifestations of this vibrational atomic motion. For example, melting occurs when the vibrations are vigorous enough to rupture large numbers of atomic bonds. A more detailed discussion of atomic vibrations and their influence on the properties of materials is presented in Chapter 19.

Microscopic Examination

4.9 BASIC CONCEPTS OF MICROSCOPY

microstructure

On occasion it is necessary or desirable to examine the structural elements and defects that influence the properties of materials. Some structural elements are of *macroscopic* dimensions; that is, they are large enough to be observed with the unaided eye. For example, the shape and average size or diameter of the grains for a polycrystalline specimen are important structural characteristics. Macroscopic grains are often evident on aluminum streetlight posts and also on highway guardrails. Relatively large grains having different textures are clearly visible on the surface of the sectioned copper ingot shown in Figure 4.13. However, in most materials the constituent grains are of *microscopic* dimensions, having diameters that may be on the order of microns,¹⁰ and their details must be investigated using some type of microscope. Grain size and shape are only two features of what is termed the **microstructure**; these and other microstructural characteristics are discussed in subsequent chapters.

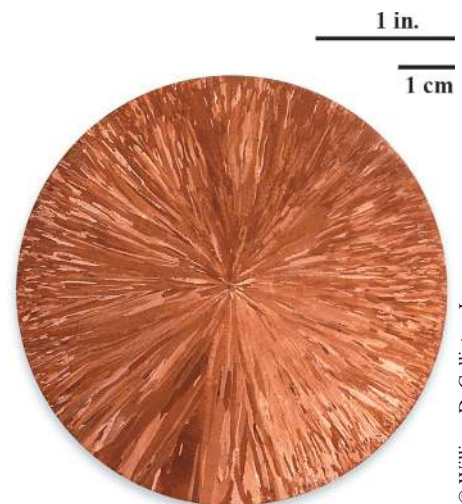
microscopy

Optical, electron, and scanning probe microscopes are commonly used in **microscopy**. These instruments aid in investigations of the microstructural features of all material types. Some of these techniques employ photographic equipment in conjunction with the microscope; the photograph on which the image is recorded is called a **photomicrograph**. In addition, many microstructural images are computer generated and/or enhanced.

photomicrograph

Microscopic examination is an extremely useful tool in the study and characterization of materials. Several important applications of microstructural examinations are as follows: to ensure that the associations between the properties and structure (and defects) are properly understood, to predict the properties of materials once these relationships have been established, to design alloys with new property combinations, to determine whether a material has been correctly heat-treated, and to ascertain the mode of mechanical fracture. Several techniques that are commonly used in such investigations are discussed next.

Figure 4.13 Cross-section of a cylindrical copper ingot. The small, needle-shape grains may be observed, which extend from the center radially outward.



¹⁰A micron (μm), sometimes called a *micrometer*, is 10^{-6} m.

4.10 MICROSCOPIC TECHNIQUES

Optical Microscopy

With *optical microscopy*, the light microscope is used to study the microstructure; optical and illumination systems are its basic elements. For materials that are opaque to visible light (all metals and many ceramics and polymers), only the surface is subject to observation, and the light microscope must be used in a reflecting mode. Contrasts in the image produced result from differences in reflectivity of the various regions of the microstructure. Investigations of this type are often termed *metallographic* because metals were first examined using this technique.

Normally, careful and meticulous surface preparations are necessary to reveal the important details of the microstructure. The specimen surface must first be ground and polished to a smooth and mirror-like finish. This is accomplished by using successively finer abrasive papers and powders. The microstructure is revealed by a surface treatment using an appropriate chemical reagent in a procedure termed *etching*. The chemical reactivity of the grains of some single-phase materials depends on crystallographic orientation. Consequently, in a polycrystalline specimen, etching characteristics vary from grain to grain. Figure 4.14*b* shows how normally incident light is reflected by three etched surface grains, each having a different orientation. Figure 4.14*a* depicts the surface structure as it might appear when viewed with the microscope; the luster or texture of each grain depends on its reflectance properties. A photomicrograph of a polycrystalline specimen exhibiting these characteristics is shown in Figure 4.14*c*.

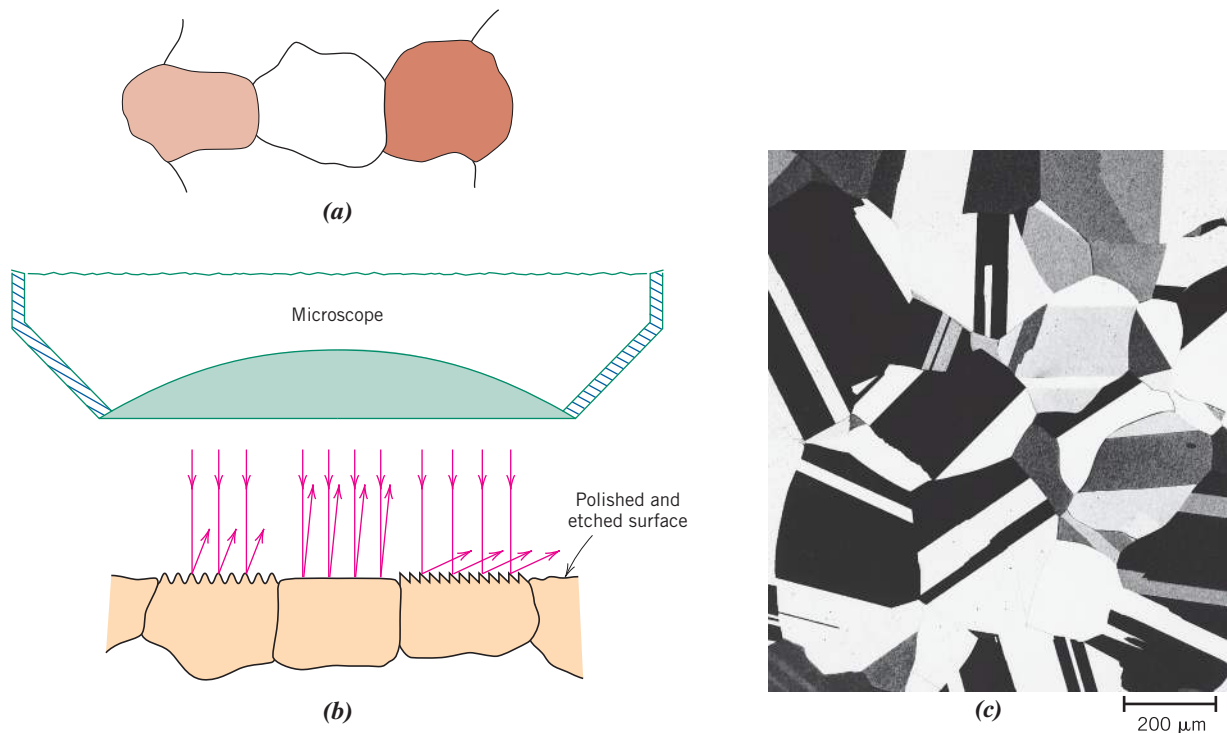


Figure 4.14 (a) Polished and etched grains as they might appear when viewed with an optical microscope. (b) Section taken through these grains showing how the etching characteristics and resulting surface texture vary from grain to grain because of differences in crystallographic orientation. (c) Photomicrograph of a polycrystalline brass specimen, 60 $\times$. [Figure (c) from J. E. Burke, *Grain Control in Industrial Metallurgy*, in “The Fundamentals of Recrystallization and Grain Growth,” Thirtieth National Metal Congress and Exposition, American Society for Metals, 1948. By permission of ASM International, Materials Park, OH. www.asmInternational.org.]

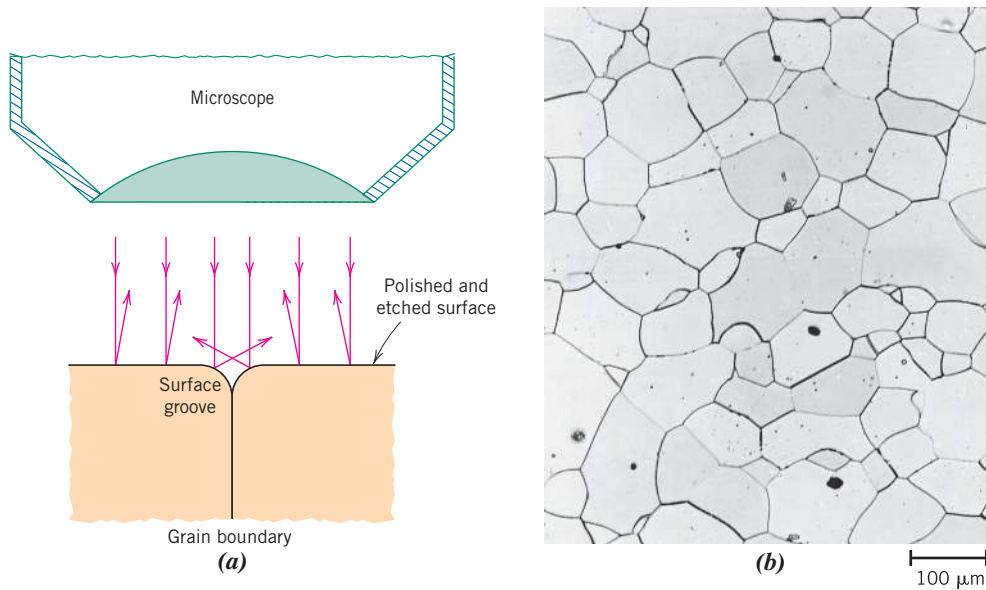


Figure 4.15 (a) Section of a grain boundary and its surface groove produced by etching; the light reflection characteristics in the vicinity of the groove are also shown. (b) Photomicrograph of the surface of a polished and etched polycrystalline specimen of an iron–chromium alloy in which the grain boundaries appear dark, 100 $\times$. [Photomicrograph courtesy of L. C. Smith and C. Brady, the National Bureau of Standards, Washington, DC (now the National Institute of Standards and Technology, Gaithersburg, MD.)]

Also, small grooves form along grain boundaries as a consequence of etching. Because atoms along grain boundary regions are more chemically active, they dissolve at a greater rate than those within the grains. These grooves become discernible when viewed under a microscope because they reflect light at an angle different from that of the grains themselves; this effect is displayed in Figure 4.15a. Figure 4.15b is a photomicrograph of a polycrystalline specimen in which the grain boundary grooves are clearly visible as dark lines.

When the microstructure of a two-phase alloy is to be examined, an etchant is often chosen that produces a different texture for each phase so that the different phases may be distinguished from each other.

Electron Microscopy

Some structural elements are too fine or small to permit observation using optical microscopy. The upper limit to the magnification possible with an optical microscope is approximately 2000 $\times$. Under such circumstances, the electron microscope, which is capable of much higher magnifications, may be employed.

An image of the structure under investigation is formed using beams of electrons instead of light radiation. According to quantum mechanics, a high-velocity electron becomes wavelike, having a wavelength that is inversely proportional to its velocity. When accelerated across large voltages, electrons can be made to have wavelengths on the order of 0.003 nm (3 pm). High magnifications and resolving powers of these microscopes are consequences of the short wavelengths of electron beams. The electron beam is focused and the image formed with magnetic lenses; otherwise, the geometry of the microscope components is essentially the same as with optical systems. Both transmission and reflection beam modes of operation are possible for electron microscopes.

Transmission Electron Microscopy

transmission electron microscope (TEM)

The image seen with a **transmission electron microscope (TEM)** is formed by an electron beam that passes through the specimen. Details of internal microstructural features are accessible to observation; contrasts in the image are produced by differences in beam scattering or diffraction produced between various elements of the microstructure or defect. Because solid materials are highly absorptive to electron beams, a specimen to be examined must be prepared in the form of a very thin foil; this ensures transmission through the specimen of an appreciable fraction of the incident beam. The transmitted beam is projected onto a fluorescent screen or a photographic film so that the image may be viewed. Magnifications approaching 1,000,000 $\times$ are possible with transmission electron microscopy, which is frequently used to study dislocations.

Scanning Electron Microscopy

scanning electron microscope (SEM)

A more recent and extremely useful investigative tool is the **scanning electron microscope (SEM)**. The surface of a specimen to be examined is scanned with an electron beam, and the reflected (or *back-scattered*) beam of electrons is collected and then displayed at the same scanning rate on a cathode ray tube (CRT; similar to a CRT television screen). The image on the screen, which may be photographed, represents the surface features of the specimen. The surface may or may not be polished and etched, but it must be electrically conductive; a very thin metallic surface coating must be applied to nonconductive materials. Magnifications ranging from 10 $\times$ to in excess of 50,000 $\times$ are possible, as are also very great depths of field. Accessory equipment permits qualitative and semiquantitative analysis of the elemental composition of very localized surface areas.

Scanning Probe Microscopy

scanning probe microscope (SPM)

In the past two decades, the field of microscopy has experienced a revolution with the development of a new family of scanning probe microscopes. The **scanning probe microscope (SPM)**, of which there are several varieties, differs from optical and electron microscopes in that neither light nor electrons are used to form an image. Rather, the microscope generates a topographical map, on an atomic scale, that is a representation of surface features and characteristics of the specimen being examined. Some of the features that differentiate the SPM from other microscopic techniques are as follows:

- Examination on the nanometer scale is possible inasmuch as magnifications as high as 10⁹ $\times$ are possible; much better resolutions are attainable than with other microscopic techniques.
- Three-dimensional magnified images are generated that provide topographical information about features of interest.
- Some SPMs may be operated in a variety of environments (e.g., vacuum, air, liquid); thus, a particular specimen may be examined in its most suitable environment.

Scanning probe microscopes employ a tiny probe with a very sharp tip that is brought into very close proximity (i.e., to within on the order of a nanometer) of the specimen surface. This probe is then raster-scanned across the plane of the surface. During scanning, the probe experiences deflections perpendicular to this plane in response to electronic or other interactions between the probe and specimen surface. The in-surface-plane and out-of-plane motions of the probe are controlled by piezoelectric (Section 18.25) ceramic components that have nanometer resolutions. Furthermore, these probe movements are monitored electronically and transferred to and stored in a computer, which then generates the three-dimensional surface image.

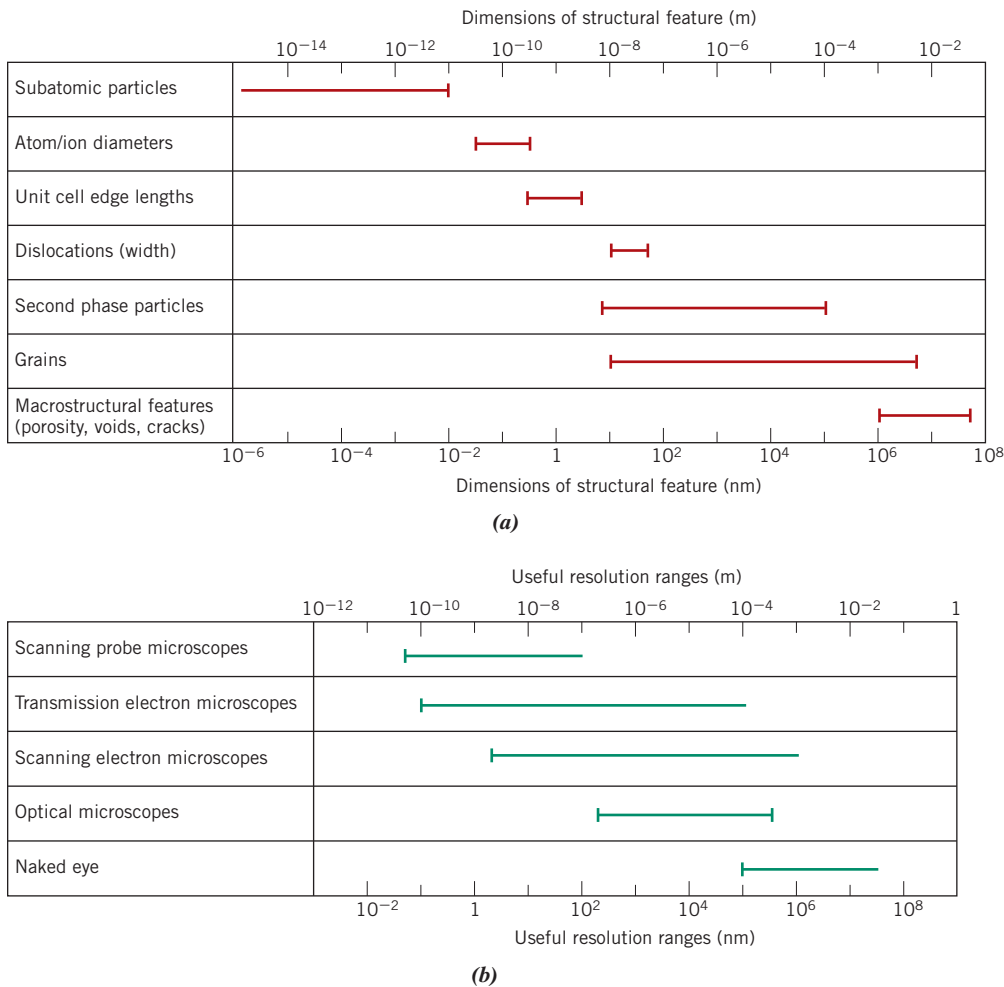


Figure 4.16 (a) Bar chart showing size ranges for several structural features found in materials. (b) Bar chart showing the useful resolution ranges for four microscopic techniques discussed in this chapter, in addition to the naked eye.

(Courtesy of Prof. Sidnei Paciornik, DCMMPUC-Rio, Rio de Janeiro, Brazil, and Prof. Carlos Pérez Bergmann, Federal University of Rio Grande do Sul, Porto Alegre, Brazil.)

These new SPMs, which allow examination of the surface of materials at the atomic and molecular level, have provided a wealth of information about a host of materials, from integrated circuit chips to biological molecules. Indeed, the advent of the SPMs has helped to usher in the era of *nanomaterials*—materials whose properties are designed by engineering atomic and molecular structures.

Figure 4.16a is a bar chart showing dimensional size ranges for several types of structures found in materials (note that the axes are scaled logarithmically). The useful dimensional resolution ranges for the several microscopic techniques discussed in this chapter (plus the naked eye) are presented in the bar chart of Figure 4.16b. For three of these techniques (SPM, TEM, and SEM), an upper resolution value is not imposed by the characteristics of the microscope and, therefore, is somewhat arbitrary and not well defined. Furthermore, by comparing Figures 4.16a and 4.16b, it is possible

to decide which microscopic technique(s) is (are) best suited for examination of each of the structure types.

4.11 GRAIN-SIZE DETERMINATION

grain size

The **grain size** is often determined when the properties of polycrystalline and single-phase materials are under consideration. In this regard, it is important to realize that for each material, the constituent grains have a variety of shapes and a distribution of sizes. Grain size may be specified in terms of average or mean grain diameter, and a number of techniques have been developed to measure this parameter.

Before the advent of the digital age, grain-size determinations were performed manually using photomicrographs. However, today, most techniques are automated and use digital images and image analyzers with the capacity to record, detect, and measure accurately features of the grain structure (i.e., grain intercept counts, grain boundary lengths, and grain areas).

We now briefly describe two common grain-size determination techniques: (1) *linear intercept*—counting numbers of grain boundary intersections by straight test lines; and (2) *comparison*—comparing grain structures with standardized charts, which are based upon grain areas (i.e., number of grains per unit area). Discussions of these techniques is from the manual perspective (using photomicrographs).

For the linear intercept method, lines are drawn randomly through several photomicrographs that show the grain structure (all taken at the same magnification). Grain boundaries intersected by all the line segments are counted. Let us represent the sum of the total number of intersections as P and the total length of all the lines by L_T . The mean intercept length $\bar{\ell}$ [in real space (at 1×—i.e., not magnified)], a measure of grain diameter, may be determined by the following expression:

$$\bar{\ell} = \frac{L_T}{PM} \quad (4.16)$$

where M is the magnification.

The comparison method of grain-size determination was devised by the American Society for Testing and Materials (ASTM).¹¹ The ASTM has prepared several standard comparison charts, all having different average grain sizes and referenced to photomicrographs taken at a magnification of 100×. To each chart is assigned a number ranging from 1 to 10, which is termed the *grain-size number*. A specimen must be prepared properly to reveal the grain structure, which is then photographed. Grain size is expressed as the grain-size number of the chart that most nearly matches the grains in the micrograph. Thus, a relatively simple and convenient visual determination of grain-size number is possible. Grain-size number is used extensively in the specification of steels.

The rationale behind the assignment of the grain-size number to these various charts is as follows: Let G represent the grain-size number, and let n be the average number of grains per square inch at a magnification of 100×. These two parameters are related to each other through the expression¹²

$$n = 2^{G-1} \quad (4.17)$$

Relationship
between ASTM
grain size number
and number of
grains per square
inch (at 100×)

¹¹ASTM Standard E112, “Standard Test Methods for Determining Average Grain Size.”

¹²Please note that in this edition, the symbol n replaces N from previous editions; also, G in Equation 4.17 is used in place of the previous n . Equation 4.17 is the standard notation currently used in the literature.

For photomicrographs taken at magnifications other than 100×, use of the following modified form of Equation 4.17 is necessary:

$$n_M \left(\frac{M}{100} \right)^2 = 2^{G-1} \quad (4.18)$$

In this expression, n_M is the number of grains per square inch at magnification M . In addition, the inclusion of the $\left(\frac{M}{100}\right)^2$ term makes use of the fact that, whereas magnification is a length parameter, area is expressed in terms of units of length squared. As a consequence, the number of grains per unit area increases with the square of the increase in magnification.

Relationships have been developed that relate mean intercept length to ASTM grain-size number; these are as follows:

$$G = -6.6457 \log \bar{\ell} - 3.298 \quad (\text{for } \bar{\ell} \text{ in mm}) \quad (4.19a)$$

$$G = -6.6353 \log \bar{\ell} - 12.6 \quad (\text{for } \bar{\ell} \text{ in in.}) \quad (4.19b)$$

At this point, it is worthwhile to discuss the representation of magnification (i.e., linear magnification) for a micrograph. Sometimes magnification is specified in the micrograph legend (e.g., “60×” for Figure 4.14b); this means the micrograph represents a 60 times enlargement of the specimen in real space. *Scale bars* are also used to express degree of magnification. A scale bar is a straight line (typically horizontal), either superimposed on or located near the micrograph image. Associated with the bar is a length, typically expressed in microns; this value represents the distance in magnified space corresponding to the scale line length. For example, in Figure 4.15b, a scale bar is located below the bottom right-hand corner of the micrograph; its “100 μm” notation indicates that 100 μm correlates with the scale bar length.

To compute magnification from a scale bar, the following procedure may be used:

1. Measure the length of the scale bar in millimeters using a ruler.
2. Convert this length into microns [i.e., multiply the value in step (1) by 1000 because there are 1000 microns in a millimeter].
3. Magnification M is equal to

$$M = \frac{\text{measured scale length (converted to microns)}}{\text{the number appearing by the scale bar (in microns)}} \quad (4.20)$$

For example, for Figure 4.15b, the measured scale length is approximately 10 mm, which is equivalent to $(10 \text{ mm})(1000 \text{ μm/mm}) = 10,000 \text{ μm}$. Inasmuch as the scale bar length is 100 μm, the magnification is equal to

$$M = \frac{10,000 \text{ μm}}{100 \text{ μm}} = 100\times$$

This is the value given in the figure legend.



Concept Check 4.4 Does the grain-size number (G of Equation 4.17) increase or decrease with decreasing grain size? Why?

[The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

EXAMPLE PROBLEM 4.5**Grain-Size Computations Using ASTM and Intercept Methods**

The following is a schematic micrograph that represents the microstructure of some hypothetical metal.

Determine the following:

- (a) Mean intercept length
- (b) ASTM grain-size number, G using Equation 4.19a

Solution

- (a) We first determine the magnification of the micrograph using Equation 4.20. The scale bar length is measured and found to be 16 mm, which is equal to 16,000 μm ; and because the scale bar number is 100 μm , the magnification is

$$M = \frac{16,000 \mu\text{m}}{100 \mu\text{m}} = 160\times$$

The following sketch is the same micrograph on which have been drawn seven straight lines (in red), which have been numbered.

The length of each line is 50 mm, and thus the total line length (L_T in Equation 4.16) is

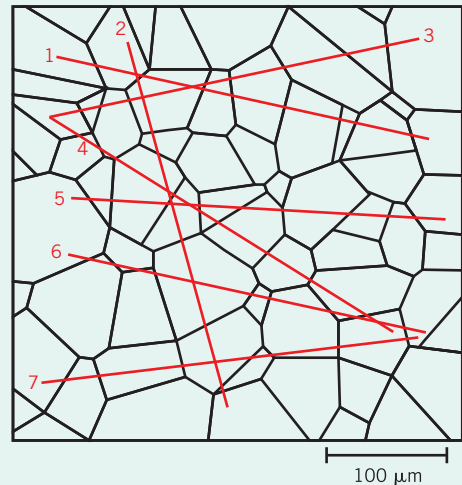
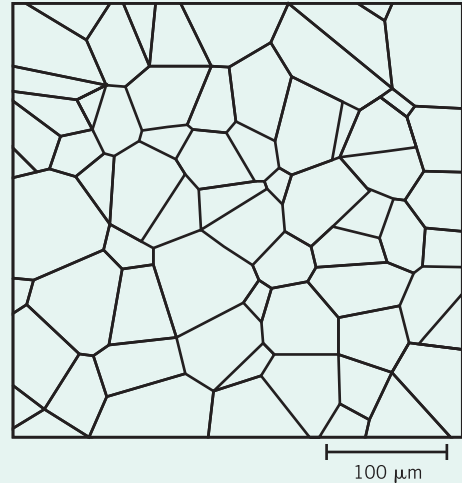
$$(7 \text{ lines})(50 \text{ mm/line}) = 350 \text{ mm}$$

Tabulated next is the number of grain-boundary intersections for each line:

Line Number	Number of Grain-Boundary Intersections
1	8
2	8
3	8
4	9
5	9
6	9
7	7
Total	58

Thus, inasmuch as $L_T = 350 \text{ mm}$, $P = 58$ grain-boundary intersections, and the magnification $M = 160\times$, the mean intercept length $\bar{\ell}$ (in millimeters in real space), Equation 4.16, is equal to

$$\begin{aligned} \bar{\ell} &= \frac{L_T}{PM} \\ &= \frac{350 \text{ mm}}{(58 \text{ grain boundary intersections})(160\times)} = 0.0377 \text{ mm} \end{aligned}$$



(b) The value of G is determined by substitution of this value for $\bar{\ell}$ into Equation 4.19a; therefore,

$$\begin{aligned} G &= -6.6457 \log \bar{\ell} - 3.298 \\ &= (-6.6457) \log(0.0377) - 3.298 \\ &= 6.16 \end{aligned}$$

SUMMARY

Vacancies and Self-Interstitials

- Point defects are those associated with one or two atomic positions; these include vacancies (or vacant lattice sites) and self-interstitials (host atoms that occupy interstitial sites).
- The equilibrium number of vacancies depends on temperature according to Equation 4.1.

Impurities in Solids

- An *alloy* is a metallic substance that is composed of two or more elements.
- A solid solution may form when impurity atoms are added to a solid, in which case the original crystal structure is retained and no new phases are formed.
- For substitutional solid solutions, impurity atoms substitute for host atoms.
- Interstitial solid solutions form for relatively small impurity atoms that occupy interstitial sites among the host atoms.
- A high degree of substitutional solid solubility of one atom type in another is possible when the Hume-Rothery rules are obeyed.

Specification of Composition

- Composition of an alloy may be specified in weight percent (on the basis of mass fraction, Equations 4.3a and 4.3b) or atom percent (on the basis of mole or atom fraction, Equations 4.5a and 4.5b).
- Expressions were provided that allow conversion of weight percent to atom percent (Equation 4.6a) and vice versa (Equation 4.7a).

Dislocations—Linear Defects

- *Dislocations* are one-dimensional crystalline defects of which there are two pure types: edge and screw.
 An *edge* may be thought of in terms of the lattice distortion along the end of an extra half-plane of atoms.
 A *screw* is as a helical planar ramp.
 For *mixed* dislocations, components of both pure edge and screw are found.
- The magnitude and direction of lattice distortion associated with a dislocation are specified by its Burgers vector.
- The relative orientations of Burgers vector and dislocation line are (1) perpendicular for edge, (2) parallel for screw, and (3) neither perpendicular nor parallel for mixed.

Interfacial Defects

- Within the vicinity of a grain boundary (which is several atomic distances wide), there is some atomic mismatch between two adjacent grains that have different crystallographic orientations.
- Across a twin boundary, atoms on one side reside in mirror-image positions of atoms on the other side.

Microscopic Techniques

- The microstructure of a material consists of defects and structural elements that are of microscopic dimensions. *Microscopy* is the observation of microstructure using some type of microscope.
- Both optical and electron microscopes are employed, usually in conjunction with photographic equipment.
- Transmissive and reflective modes are possible for each microscope type; preference is dictated by the nature of the specimen as well as the structural element or defect to be examined.
- The two types of electron microscopes are transmission (TEM) and scanning (SEM). For TEM, an image is formed from an electron beam that, although passing through the specimen, is scattered and/or diffracted. SEM employs an electron beam that raster-scans the specimen surface; an image is produced from back-scattered or reflected electrons.
- A scanning probe microscope employs a small and sharp-tipped probe that raster-scans the specimen surface. A computer-generated and three-dimensional image of the surface results having nanometer resolution.

Grain-Size Determination

- With the intercept method used to measure grain size, a series of straight-line segments are drawn on the photomicrograph. The number of grain boundaries that are intersected by these lines are counted, and the *mean intercept length* (a measure of grain diameter) is computed using Equation 4.16.
- Comparison of a photomicrograph (taken at a magnification of 100×) with ASTM standard comparison charts may be used to specify grain size in terms of a grain-size number.

Equation Summary

Equation Number	Equation	Solving For
4.1	$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$	Number of vacancies per unit volume
4.2	$N = \frac{N_A \rho}{A}$	Number of atomic sites per unit volume
4.3a	$C_1 = \frac{m_1}{m_1 + m_2} \times 100$	Composition in weight percent
4.5a	$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$	Composition in atom percent
4.6a	$C'_1 = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100$	Conversion from weight percent to atom percent
4.7a	$C_1 = \frac{C'_1 A_1}{C'_1 A_1 + C'_2 A_2} \times 100$	Conversion from atom percent to weight percent
4.9a	$C''_1 = \left(\frac{C_1}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3$	Conversion from weight percent to mass per unit volume
4.10a	$\rho_{\text{ave}} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}$	Average density of a two-component alloy

(continued)

Equation Number	Equation	Solving For
4.11a	$A_{\text{ave}} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}}$	Average atomic weight of a two-component alloy
4.16	$\bar{\ell} = \frac{L_T}{PM}$	Mean intercept length (measure of average grain diameter)
4.17	$n = 2^{G-1}$	Number of grains per square inch at a magnification of 100×
4.18	$n_M = (2^{G-1}) \left(\frac{100}{M} \right)^2$	Number of grains per square inch at a magnification other than 100×

List of Symbols

Symbol	Meaning
A	Atomic weight
G	ASTM grain-size number
k	Boltzmann's constant (1.38×10^{-23} J/atom·K, 8.62×10^{-5} eV/atom·K)
L_T	Total line length (intercept technique)
M	Magnification
m_1, m_2	Masses of elements 1 and 2 in an alloy
N_A	Avogadro's number (6.022×10^{23} atoms/mol)
n_{m1}, n_{m2}	Number of moles of elements 1 and 2 in an alloy
P	Number of grain boundary intersections
Q_v	Energy required for the formation of a vacancy
ρ	Density

Important Terms and Concepts

alloy	imperfection	screw dislocation
atomic vibration	interstitial solid solution	self-interstitial
atom percent	microscopy	solid solution
Boltzmann's constant	microstructure	solute
Burgers vector	mixed dislocation	solvent
composition	photomicrograph	substitutional solid solution
dislocation line	point defect	transmission electron microscope (TEM)
edge dislocation	scanning electron microscope (SEM)	vacancy
grain size	scanning probe microscope (SPM)	weight percent

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